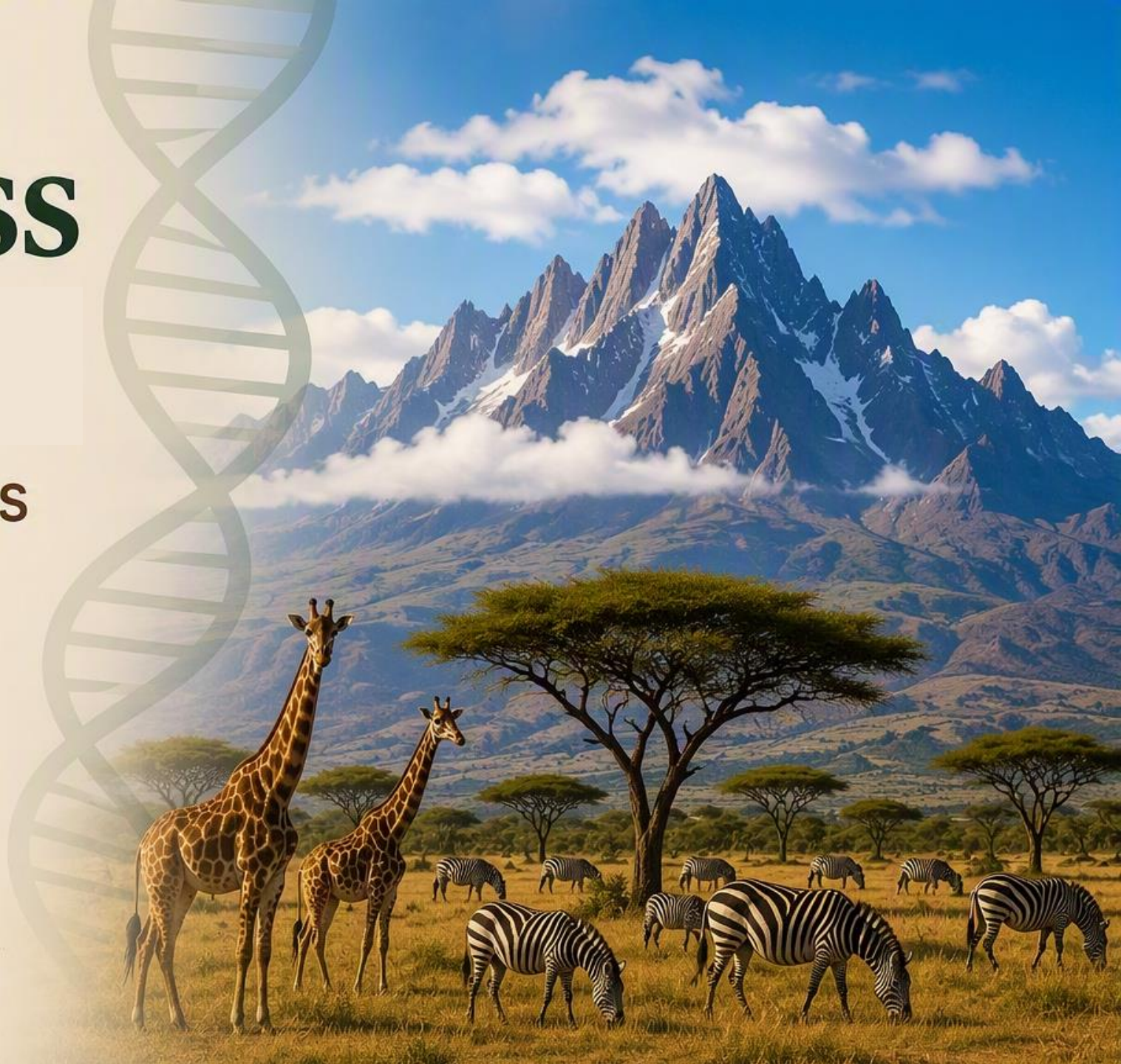
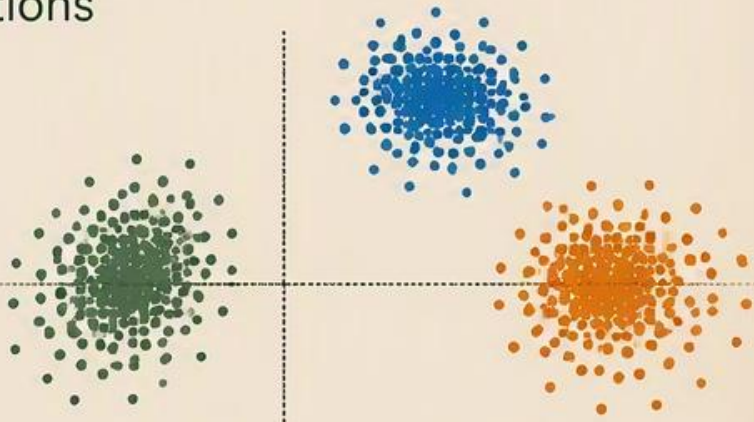


Relatedness

IN POPULATION GENETICS

Understanding genetic similarity
within and differentiation between
populations



Today's class

Both methods compare observed allele sharing to expected sharing under a demographic model.

Relatedness

How related are two individuals?
IBD sharing vs expectation under
random mating

Relatedness asks: how much more do two individuals share than expected for random individuals?

FST

How differentiated are
populations?
Allele frequency variance among
groups

FST asks: how much do allele frequencies differ among populations compared to within?

Relatedness

Relatedness

How related are two individuals?
IBD sharing vs expectation under
random mating

Relatedness asks: how much more do two individuals share than expected for random individuals?

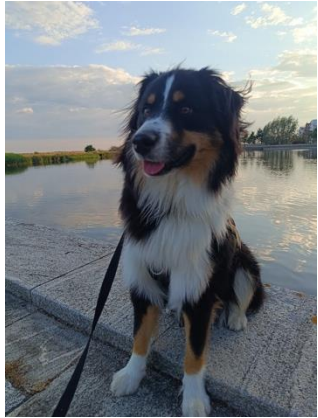
Background:

- What is IBD sharing
- **Inbreeding** and relatedness
- Different relatedness coefficients
- Different methods to measure relatedness

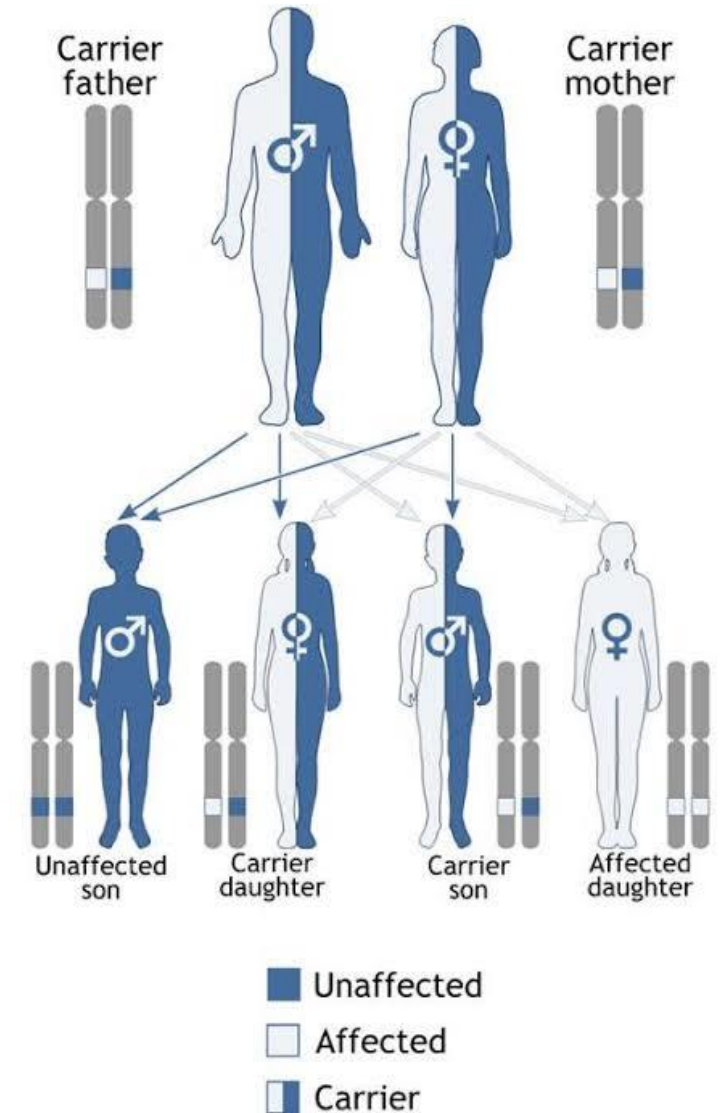
Aims:

- Understand that IBD is not just a number but a pattern across the genome
- How the different coefficients are related
- How relatedness can influence analysis

Relatedness – broad view

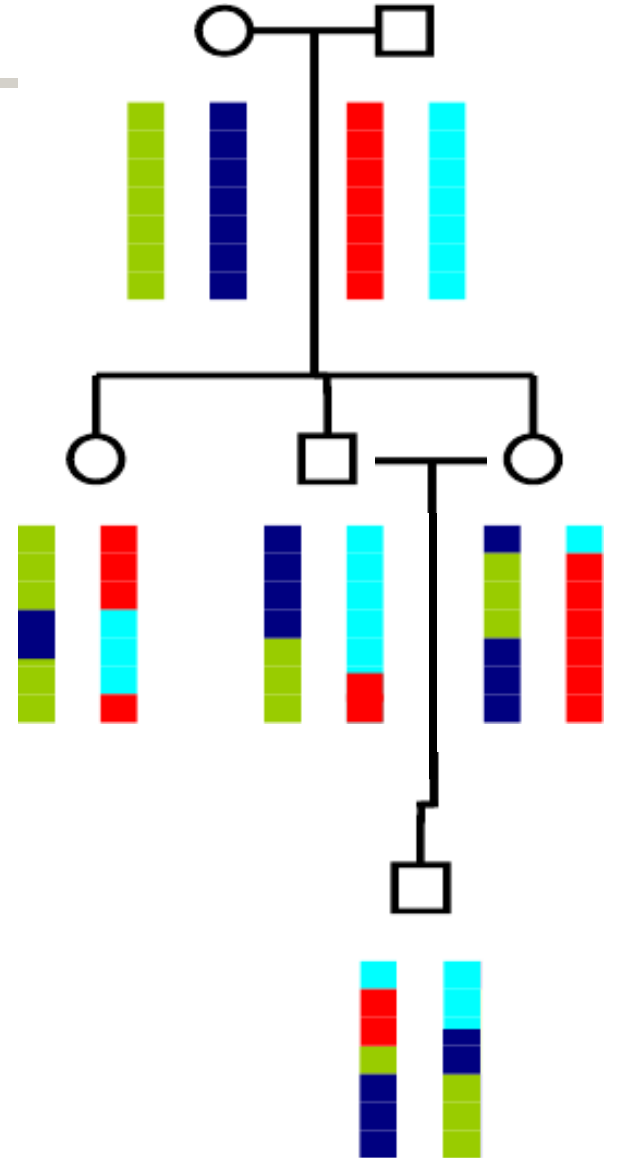


- Meaningful resemblance
- Genes transmission
 - Characteristics
 - Diseases



Not so simple

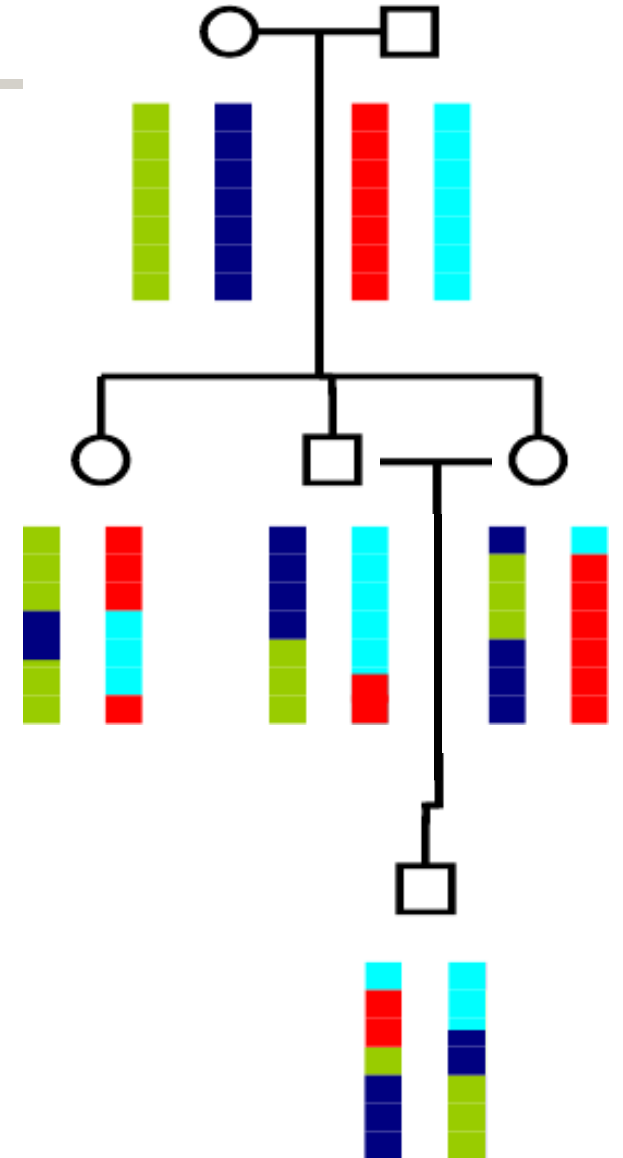
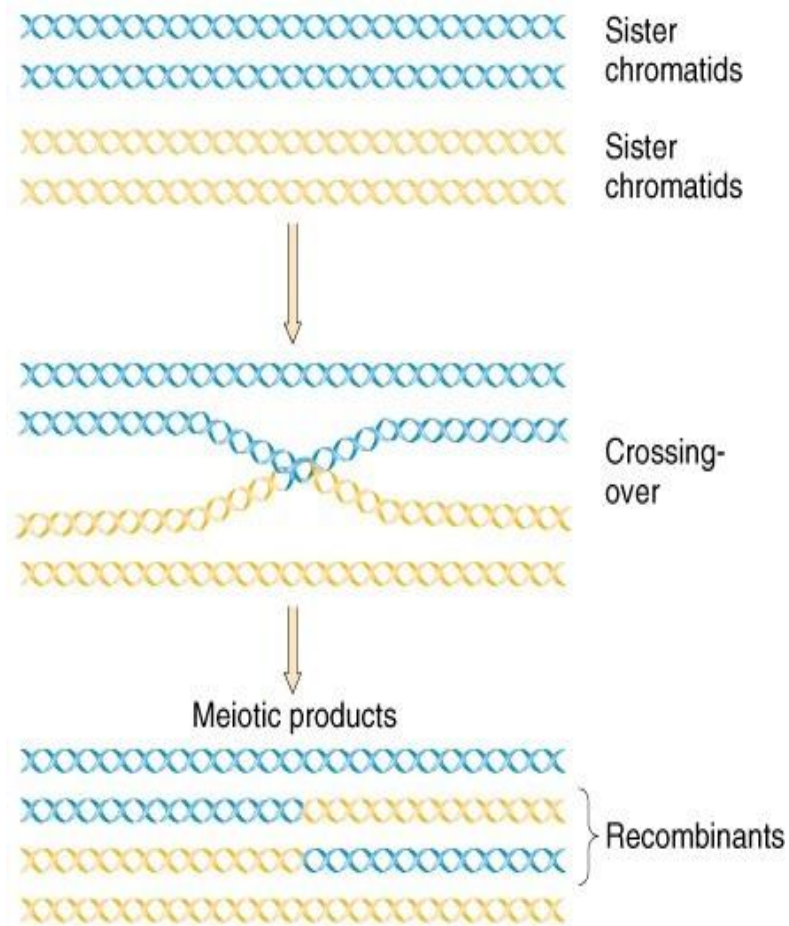
Relatedness between a pair of individuals is usually described using the concept of identity-by-descent (IBD), which is genetic identity due to recent common ancestry.



Not so simple

Relatedness between a pair of individuals is usually described using the concept of identity-by-descent (IBD), which is genetic identity due to recent common ancestry.

Because of a finite number of recombination events we expect the IBD state of a loci to be dependent of the state of adjacent loci



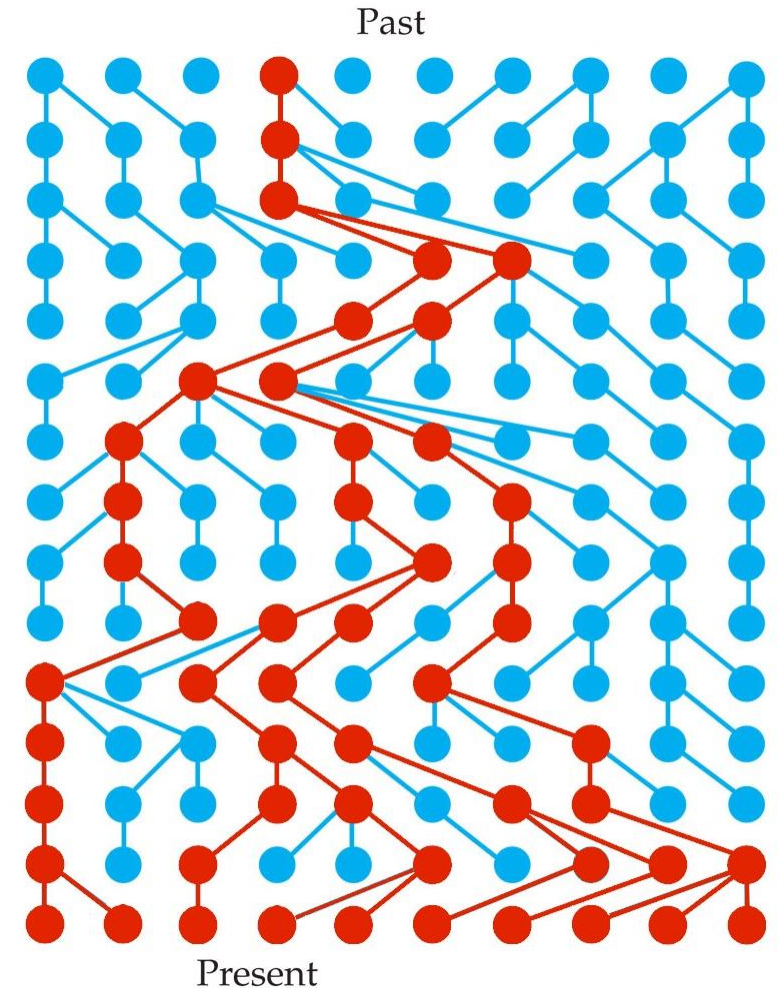
Not so simple

Relatedness between a pair of individuals is usually described using the concept of identity-by-descent (IBD), which is genetic identity due to recent common ancestry.

Because of a finite number of recombination events we expect the IBD state of a loci to be dependent of the state of adjacent loci

All of the individuals in a population are related to each other by a giant pedigree (family tree)

All individuals are related to one another by varying levels of relatedness



IBD vs IBS: Same Genotype, Different Story

Identity by State (IBS)

Two unrelated individuals from different populations can share the same allele (IBS) without being related (IBD).

Individual A



Individual B



*Same alleles (A1, A2) but
different ancestral copies*

Identity by Descent (IBD)

Two alleles are IBD if they are copies of the **same ancestral allele** transmitted through pedigree pathways.

Individual A



Individual B

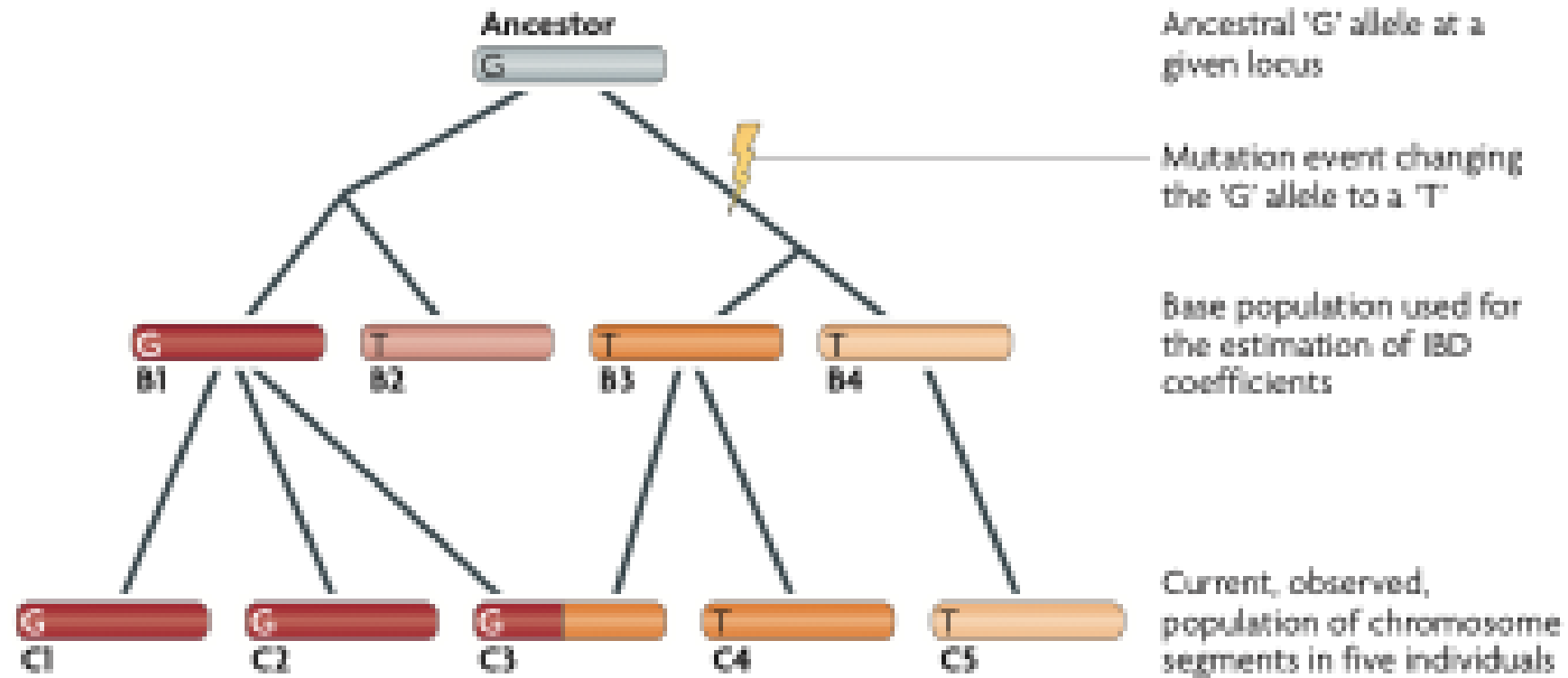


*shared
ancestor*

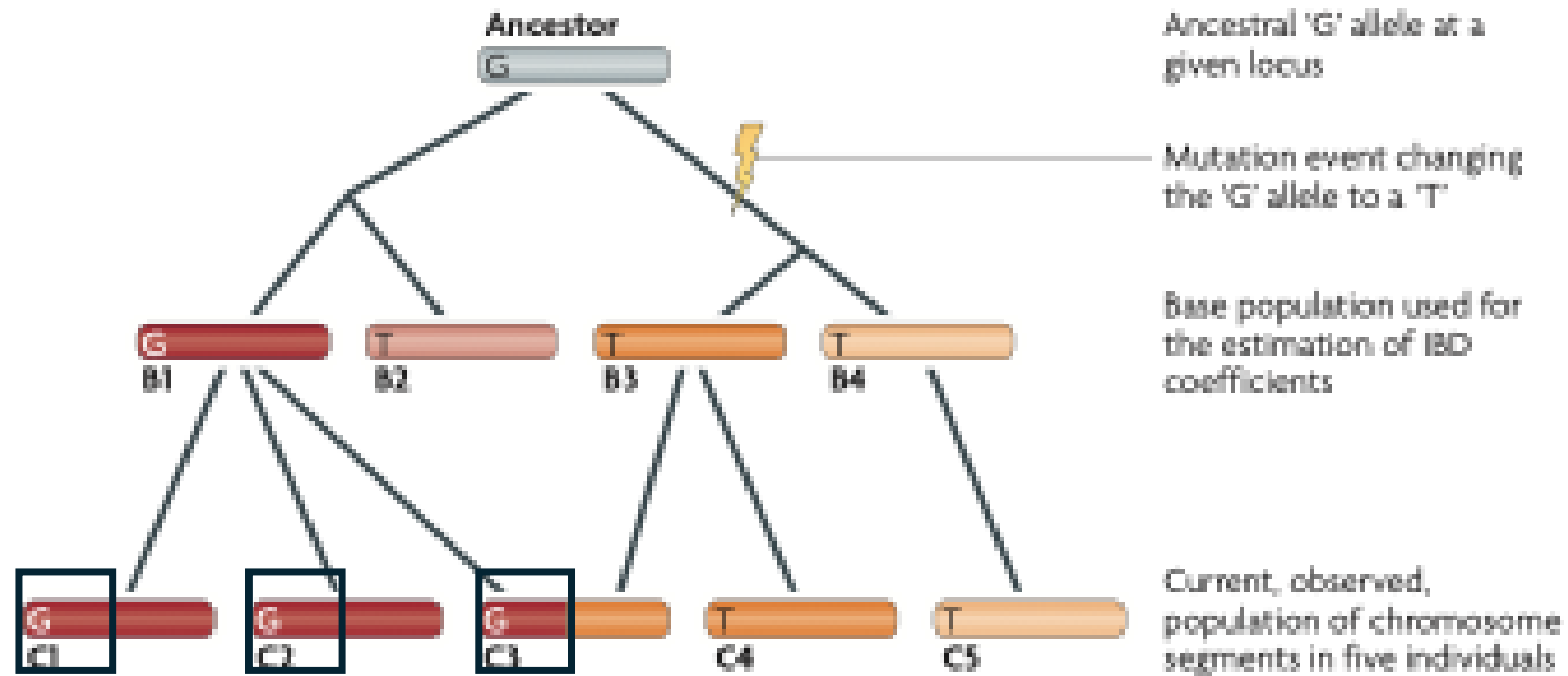
*Same alleles AND inherited from
a common ancestor*

Two alleles are IBS if they carry the same nucleotide—regardless of ancestry. IBS includes IBD, but not vice versa.

Specific example



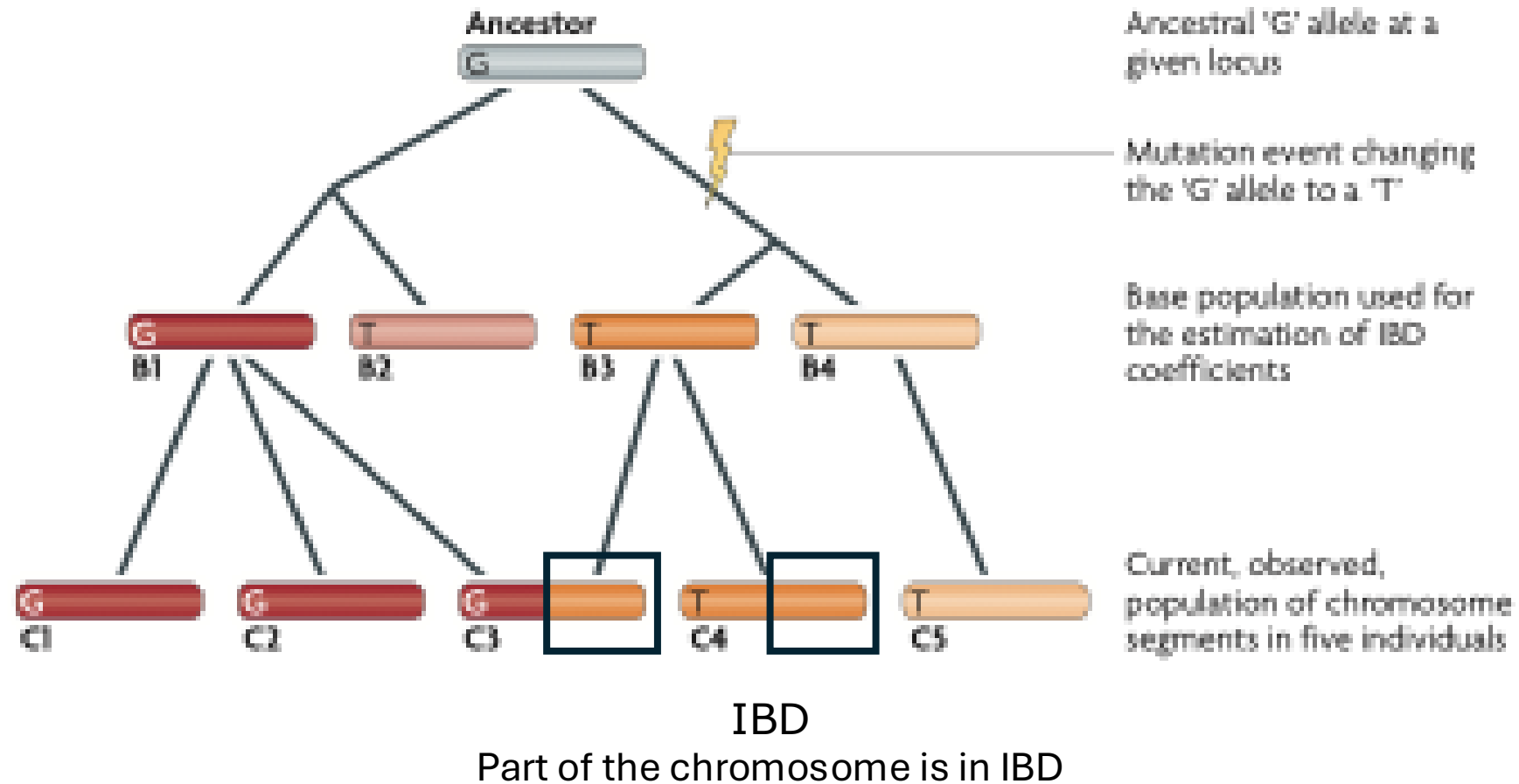
Specific example



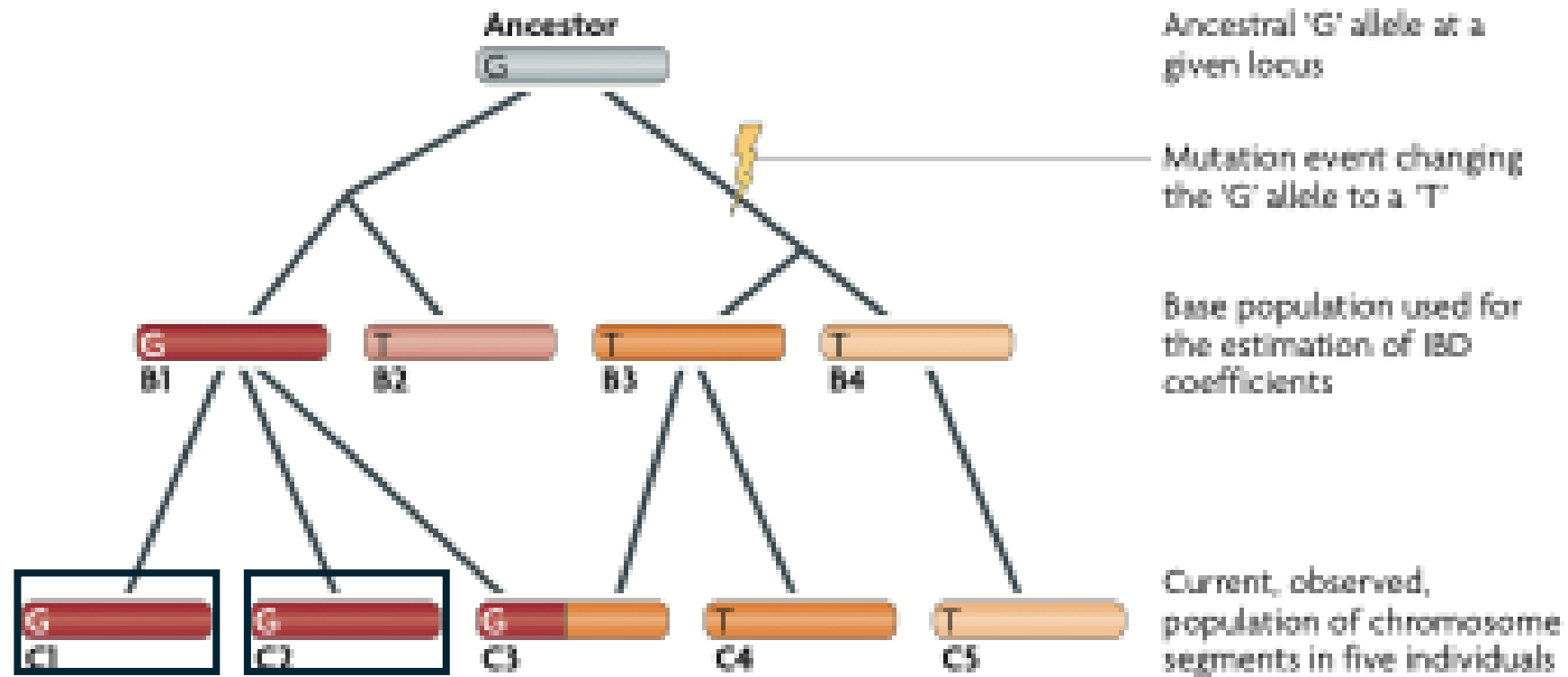
IBD

The **G** alleles in **C1**, **C2** and **C3** are IBD to each other as all three descend from the G allele in B1

Specific example



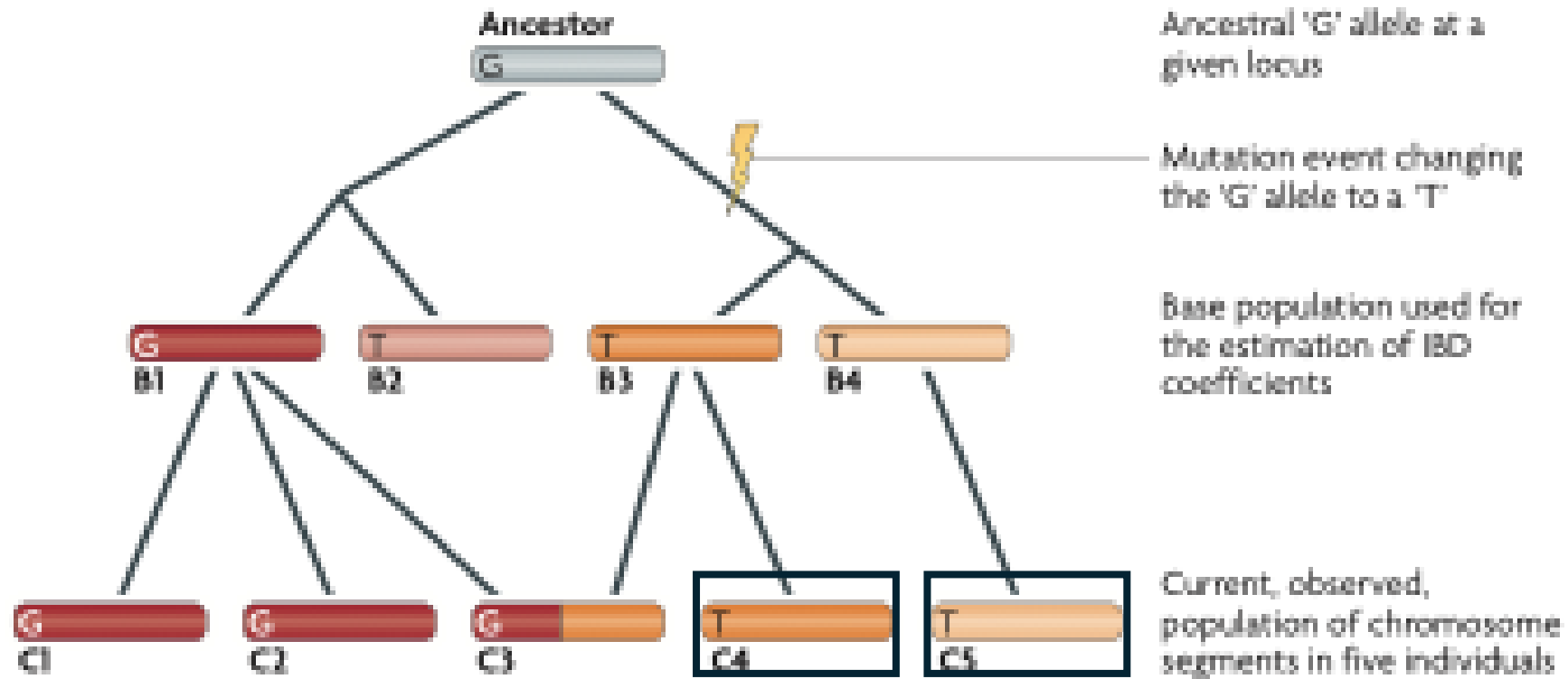
Specific example



IBD

The **whole chromosome** in **C1** and **C2** are IBD because they descend from B1 without recombination

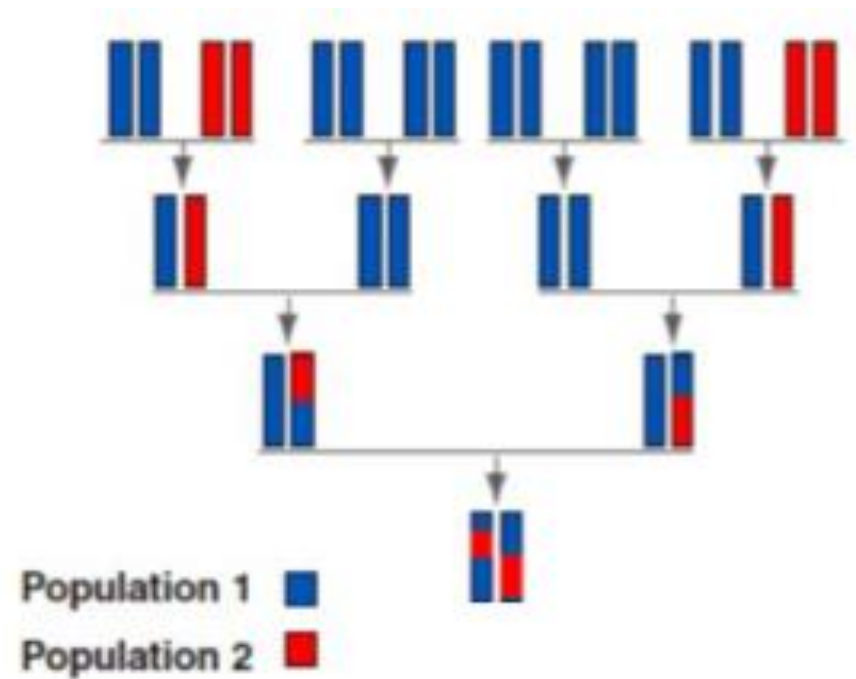
Specific example



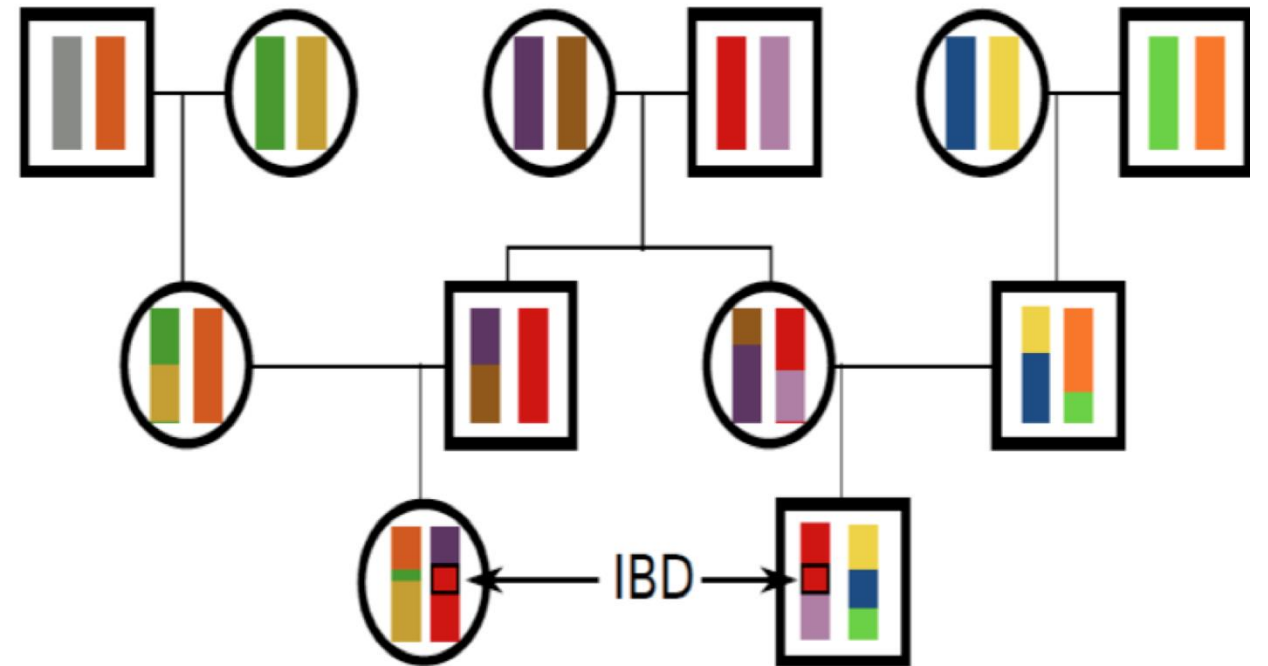
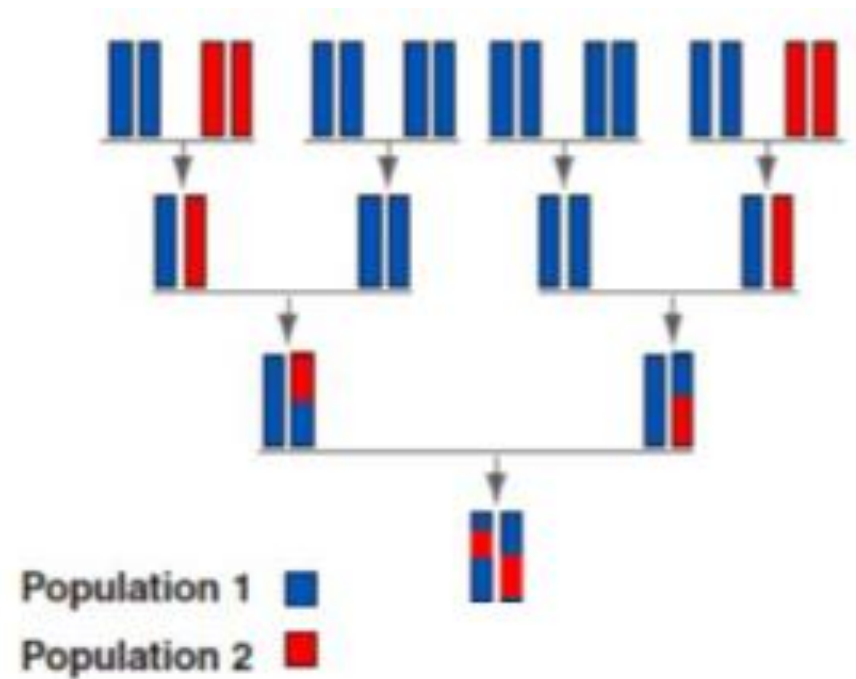
IBS not IBD

As the chromosomes descend from **different** ancestors

Pedigree



IBD - Identity by Descent

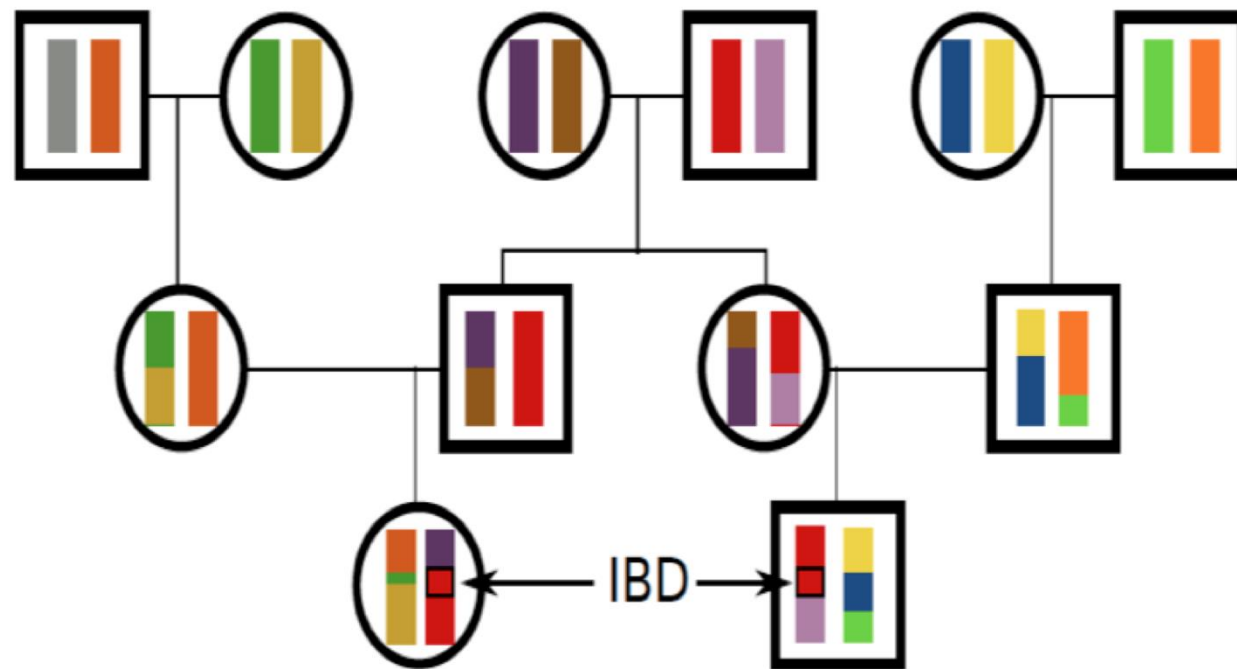
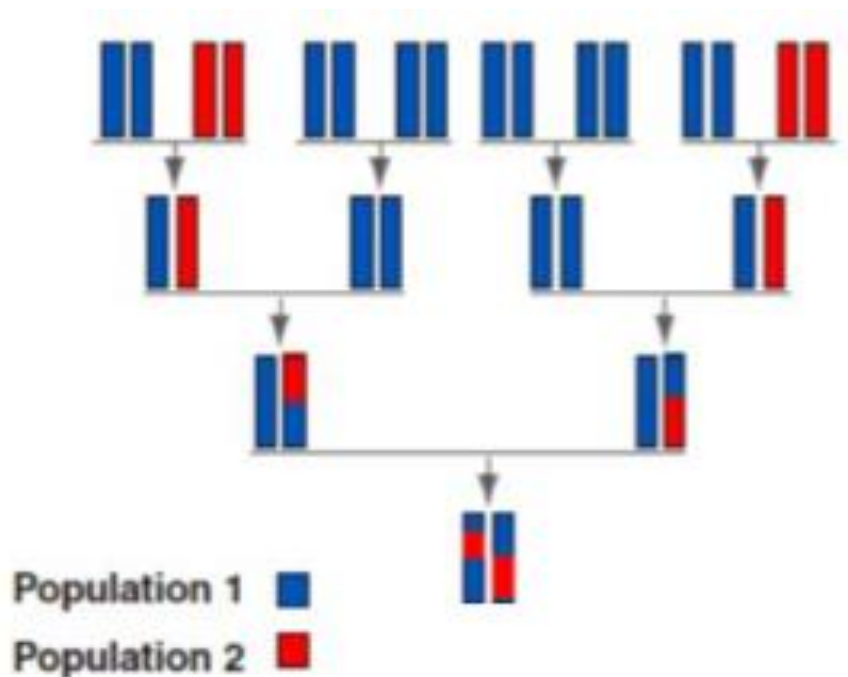


IBD - Identity by Descent

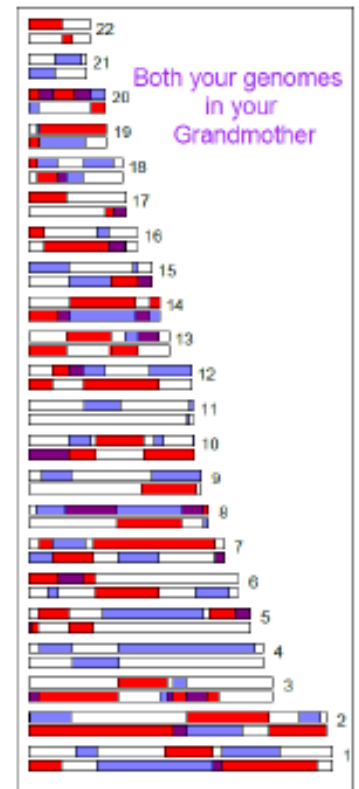
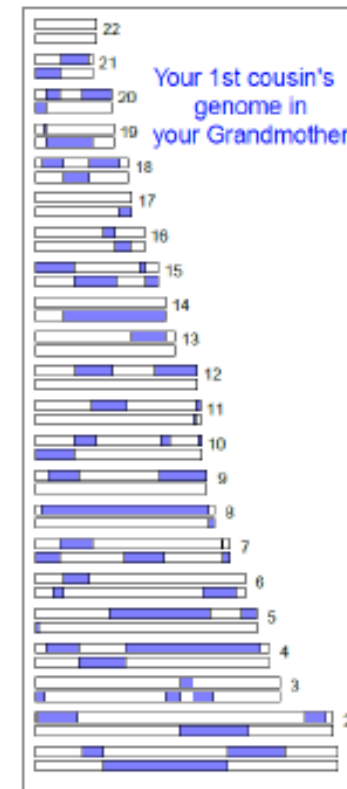
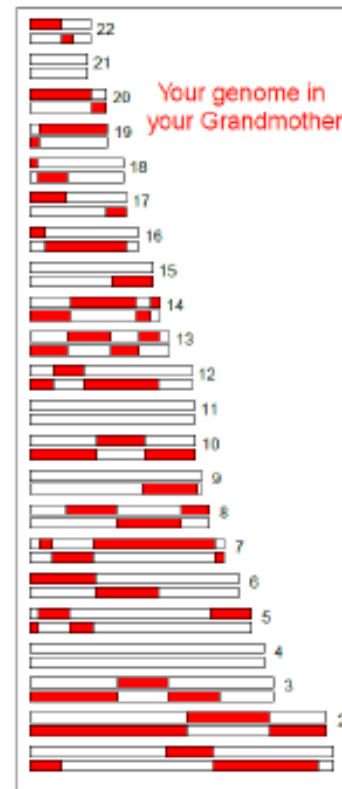
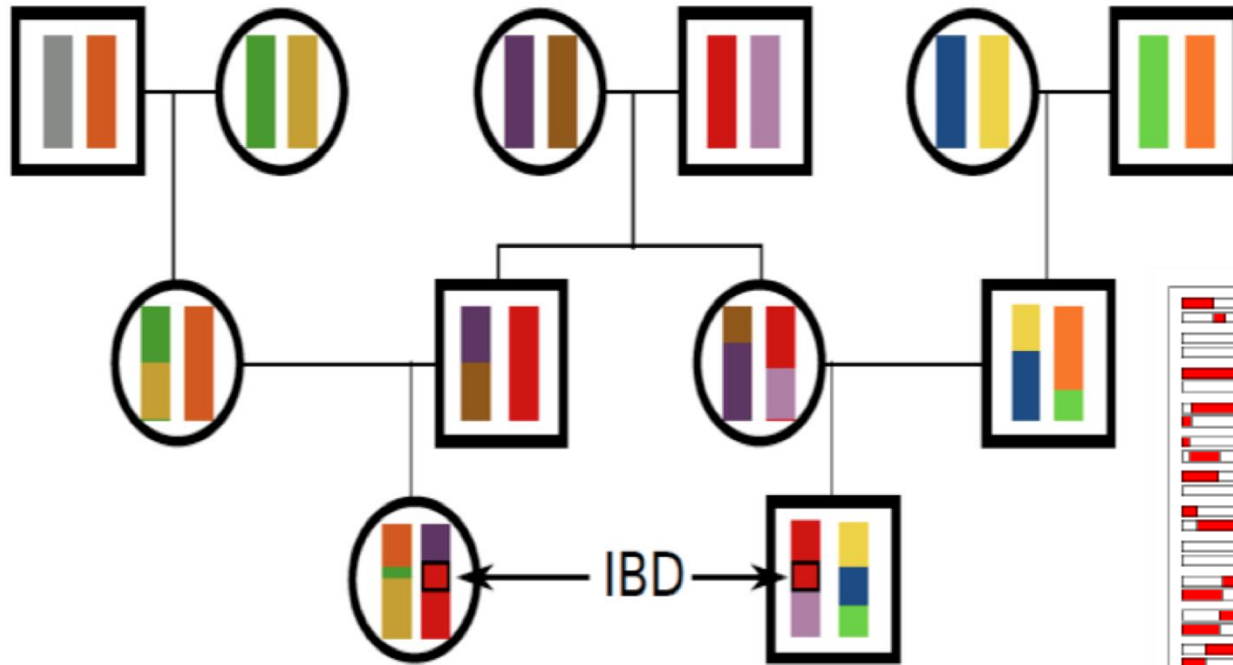
We will define two alleles to be identical by descent (IBD) if they are identical due to transmission from a common ancestor in the past few generations.

The IBD leads to excess identity compared to what would be expected given population frequency.

They occur in consecutive regions

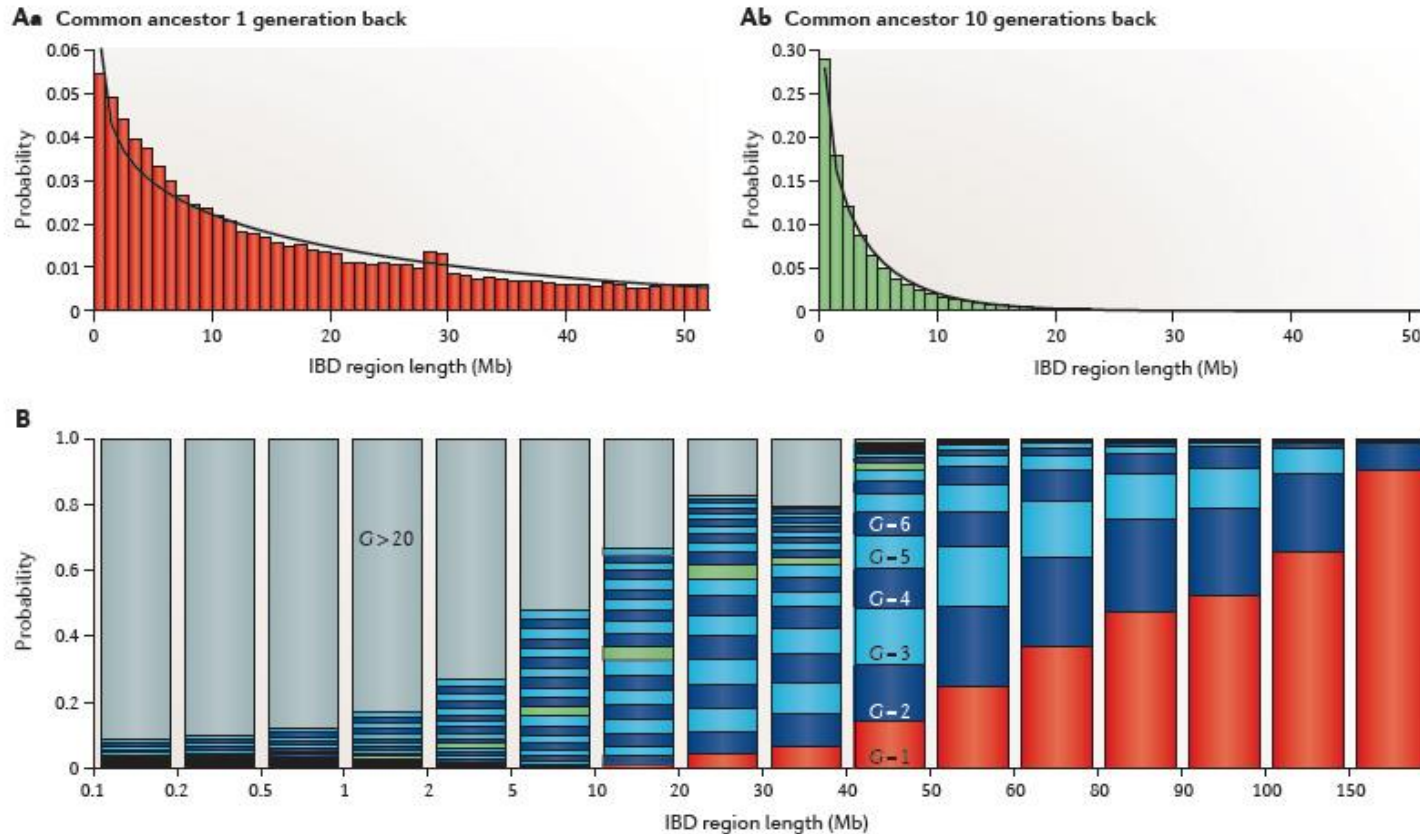


IBD - Identity by Descent



Calculating IBD

What lengths do IBD segments have for a known ancestor age?



Given an IBD segment of this length, how probable is each ancestor generation?

A recent common ancestor tends to produce **long IBD segments**, whereas an older common ancestor produces **shorter segments**.

How many generations back should I pick?

The shortest IBD segment length you can call reliably.

We stick with **recent shared ancestry**, because with deeper times it gets increasingly hard to distinguish from IBS.

IBD States: IBD0, IBD1, IBD2

For any pair of diploid individuals, at each locus they share 0, 1, or 2 alleles IBD

IBD0

No alleles shared IBD

Individual A

A1   Alele2

A1   Alele2

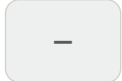
Individual B

Unrelated

IBD1

One allele shared IBD

Individual A

A1   Alele2

A1   Alele2

Individual B

Parent-offspring

IBD2

Both alleles shared IBD

Individual A

A1   Alele2

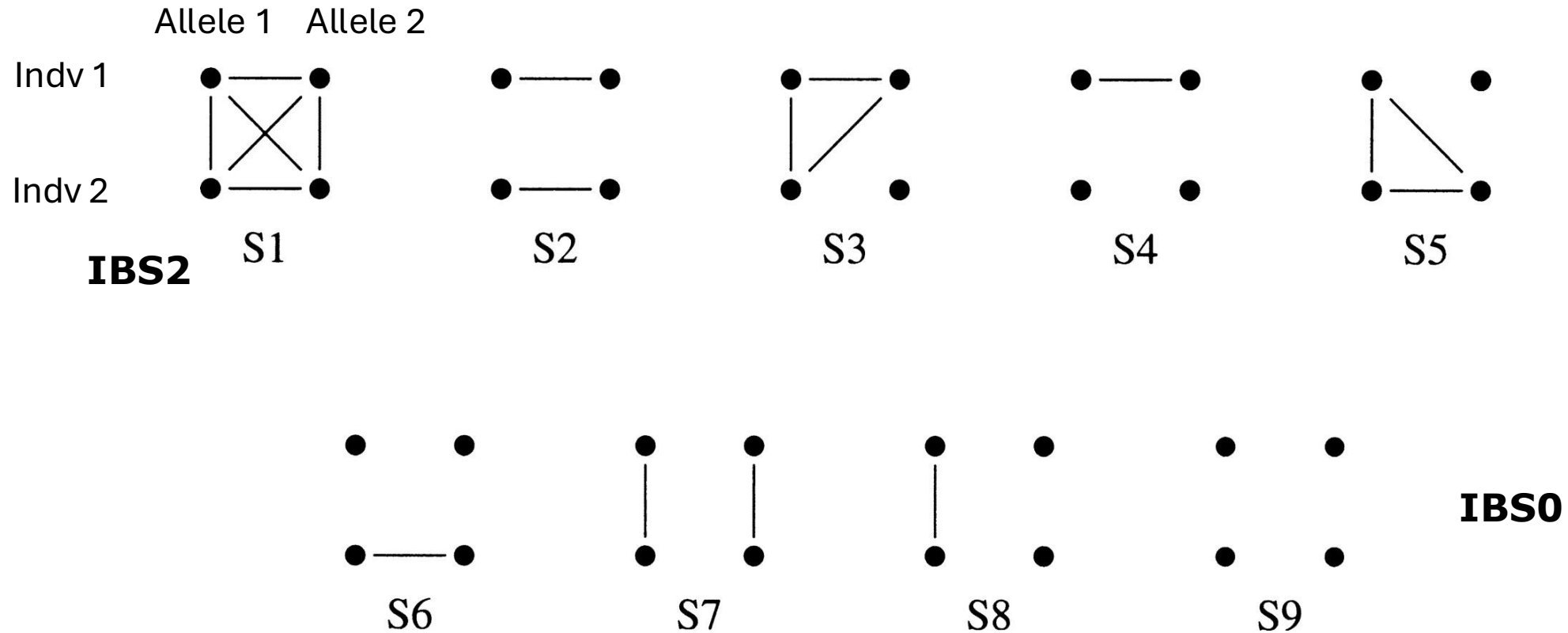
A1   Alele2

Individual B

MZ twins / inbred

Different IBD stages

Given two diploid individuals, there are 9 identity by descent models available between them



Calculate relatedness

How much a pair shares IBD is quantified in several ways :

$R = (k_0, k_1, k_2)$: fractions of genome with 0, 1 and 2 alleles IBD (**IBD states distribution**)

k_0 – probability of sharing **no** IBD alleles at a locus

k_1 – probability of sharing **one** IBD alleles

k_2 – probability of sharing **two** IBD alleles

$$k_0 + k_1 + k_2 = 1$$

r (**coefficient of relatedness**): expected fraction of alleles of shared IBD

$$r = \frac{1}{2}k_1 + k_2 \Rightarrow r = 0 \times k_0 + \frac{1}{2}k_1 + 1 \times k_2$$

ϕ (**kinship coefficient**): probability that 2 random alleles are IBD ($\phi = 0.5r$)

$$\phi = \frac{1}{4}k_1 + \frac{1}{2}k_2$$

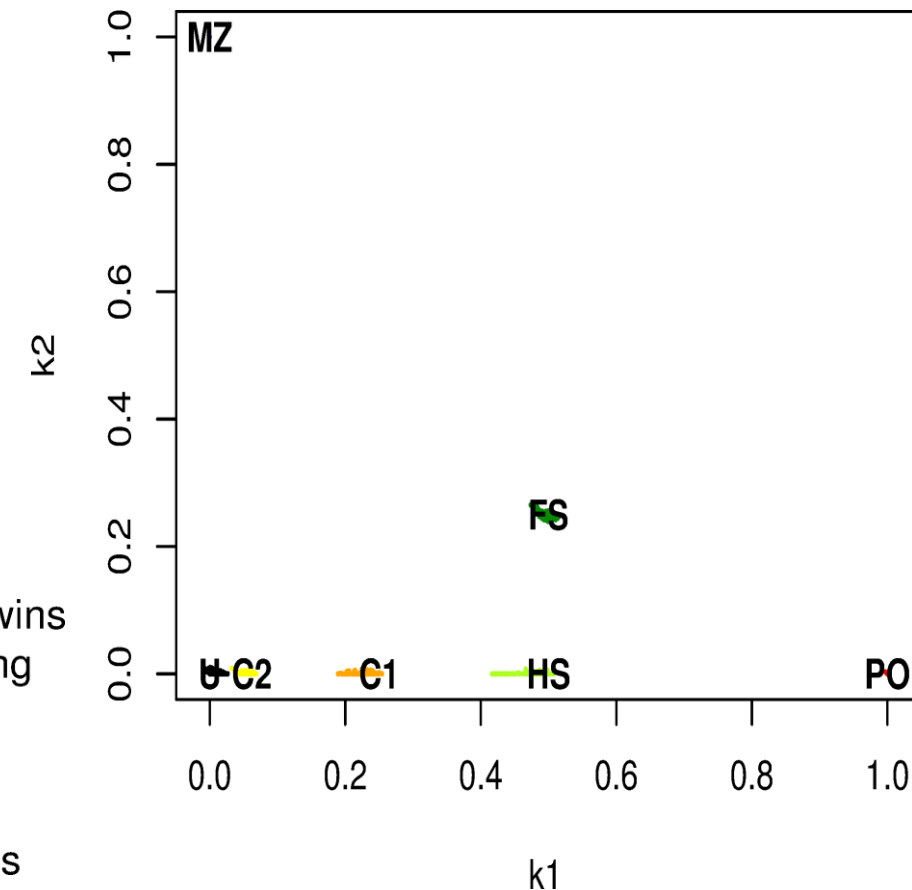
Inference of relationships

We expect different relationships to have different values of R , e.g.:

Relationship	k_0	k_1	k_2
Monozygotic twins	0	0	1
Parent-offspring	0	1	0
Siblings	0.25	0.5	0.25
Halfsiblings/Uncle-nephew/grandparent-child	0.5	0.5	0
First cousins	0.75	0.25	0
Second cousins	0.9375	0.0625	0
Unrelated	1	0	0

Hence we can (often) use R to infer how two individuals are related

- Unrelated
- Monozygotic twins
- Parent-offspring
- Full siblings
- Half siblings
- Cousins
- Second cousins



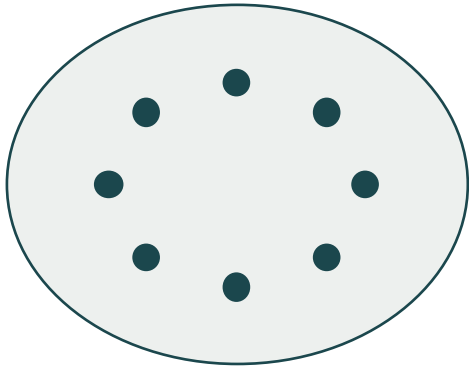
All together

Relationship	k_0	k_1	k_2	r	ϕ
Identical twins / Self	0	0	1	1	0.5
Parent-offspring	0	1	0	0.5	0.25
Full siblings	1/4	1/2	1/4	0.5	0.25
Half siblings	1/2	1/2	0	0.25	0.125
First cousins				0.125	0.0625
Unrelated	1	0	0	0	0

$$(k_0, k_1, k_2) \rightarrow \phi = \frac{1}{4}k_1 + \frac{1}{2}k_2 \rightarrow r = 2\phi = \frac{1}{2}k_1 + k_2$$

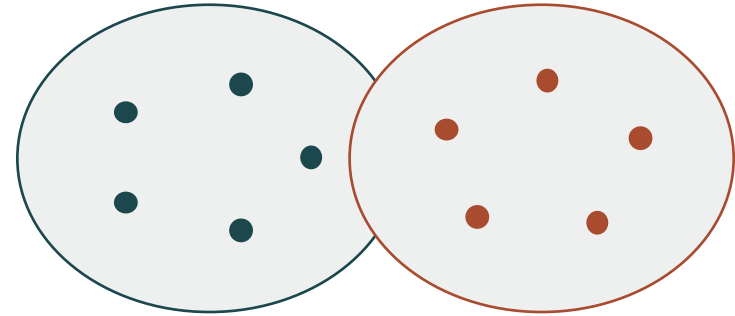
Population structure confounds Relatedness

Scenario A: Single Population



*Shared alleles reflect
IBD (recent common ancestry)*

Scenario B: Structured Populations

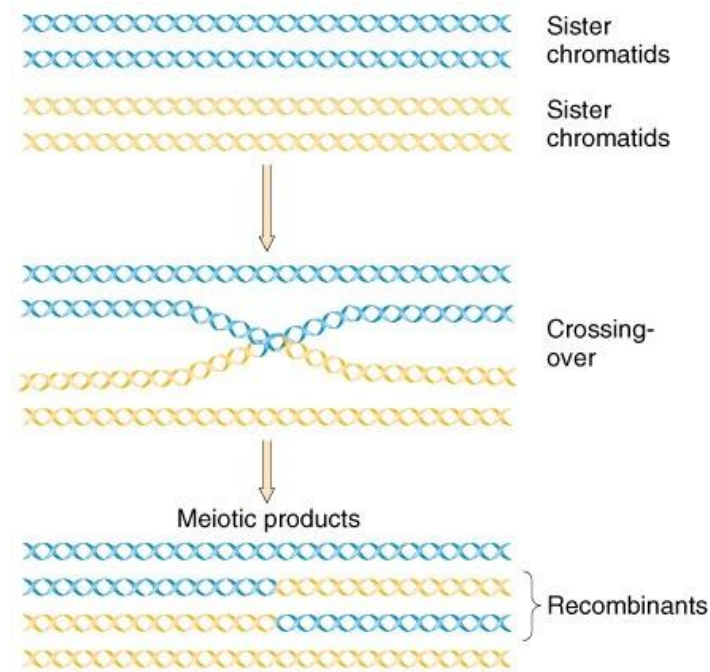
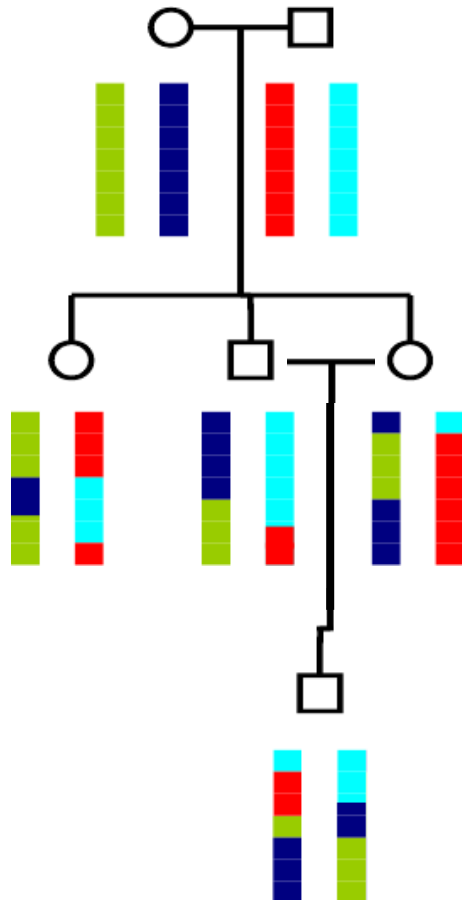


*Shared alleles reflect
population ancestry, not IBD!*

Problem: Traditional relatedness estimators use population allele frequencies. When structure exists, individuals from the same sub-population appear "related" even when they are not.

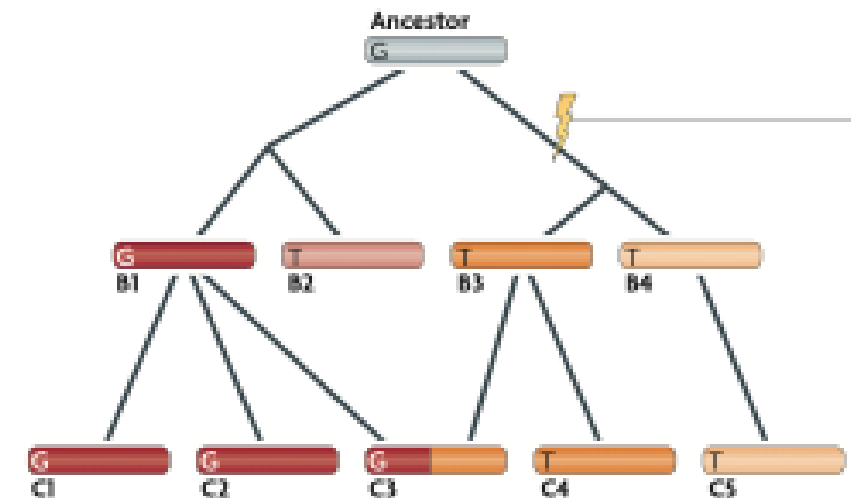
A definition for IBD

Two alleles are *Identical by descent* (IBD) if they are direct copies of a single ancestral allele



Because of a finite number of recombination events we expect the IBD state of a loci to be dependent of the state of adjacent loci

Population structure matters for IBD calling



Relatedness is important

1

Analysis

Many analysis assume that all individuals analyzed are unrelated and if related individuals are not removed or controlled for they can lead to highly inflated false-positive rates.

2

Diseases

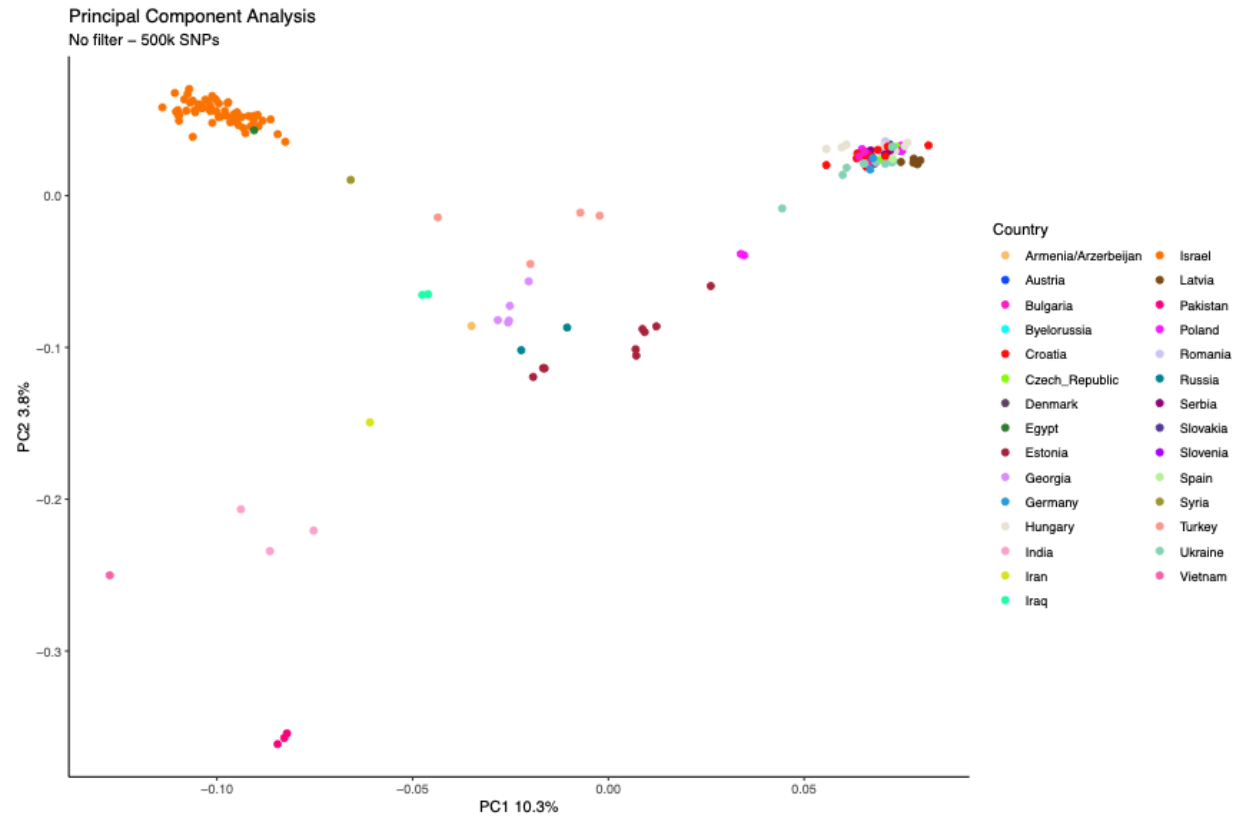
3

Inbreeding

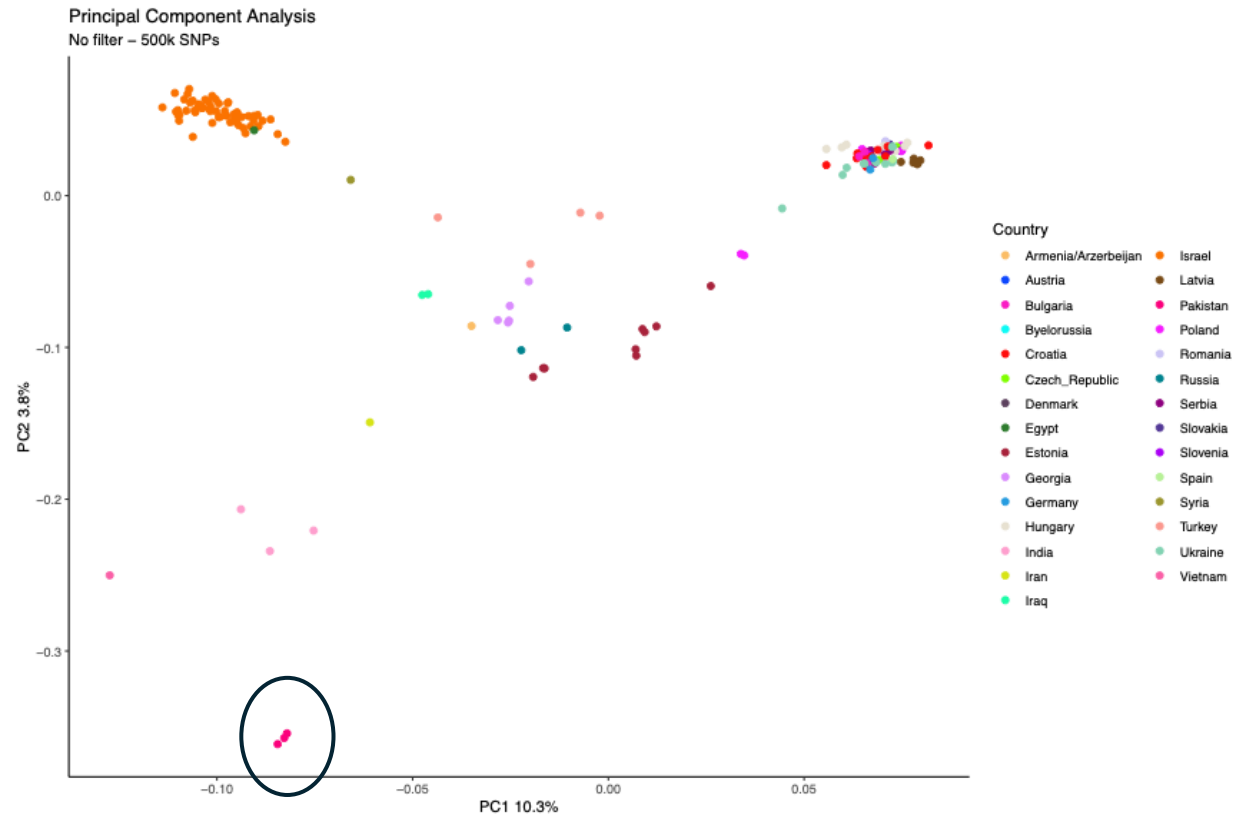
4

Conservation

PCA results – population structure

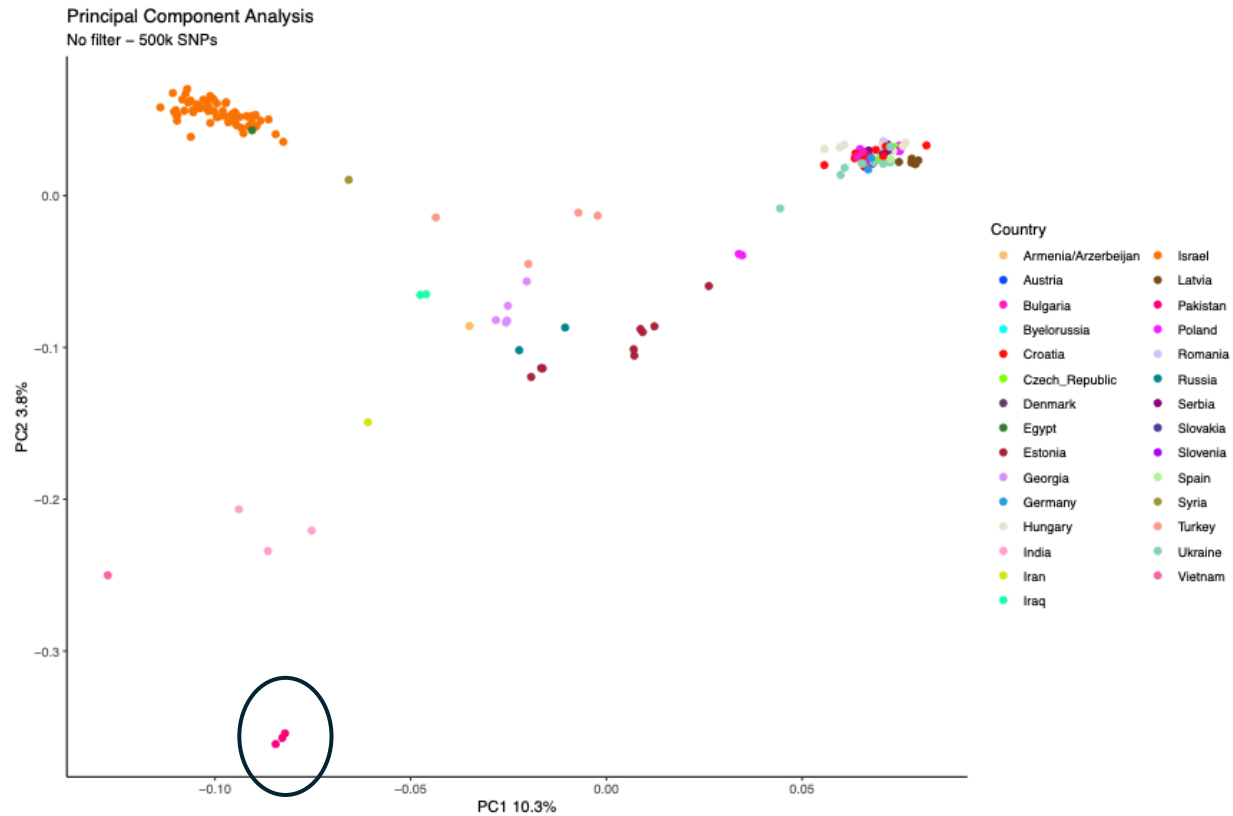


PCA results – population structure



Pakistan

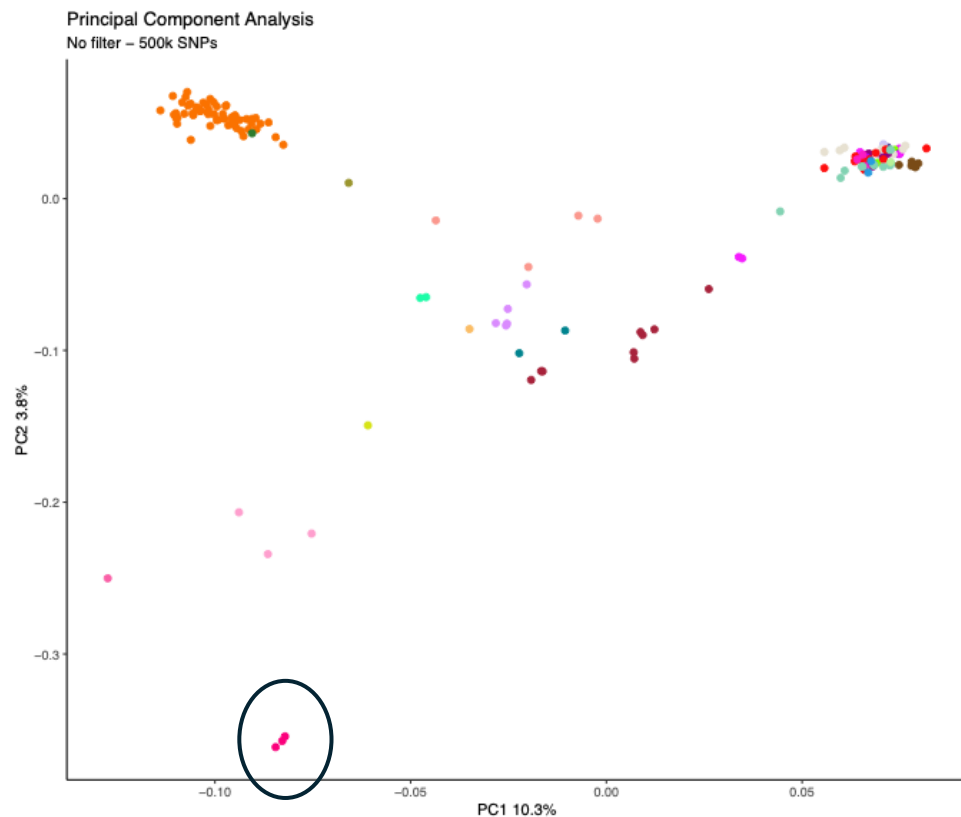
PCA results – population structure



Pakistan

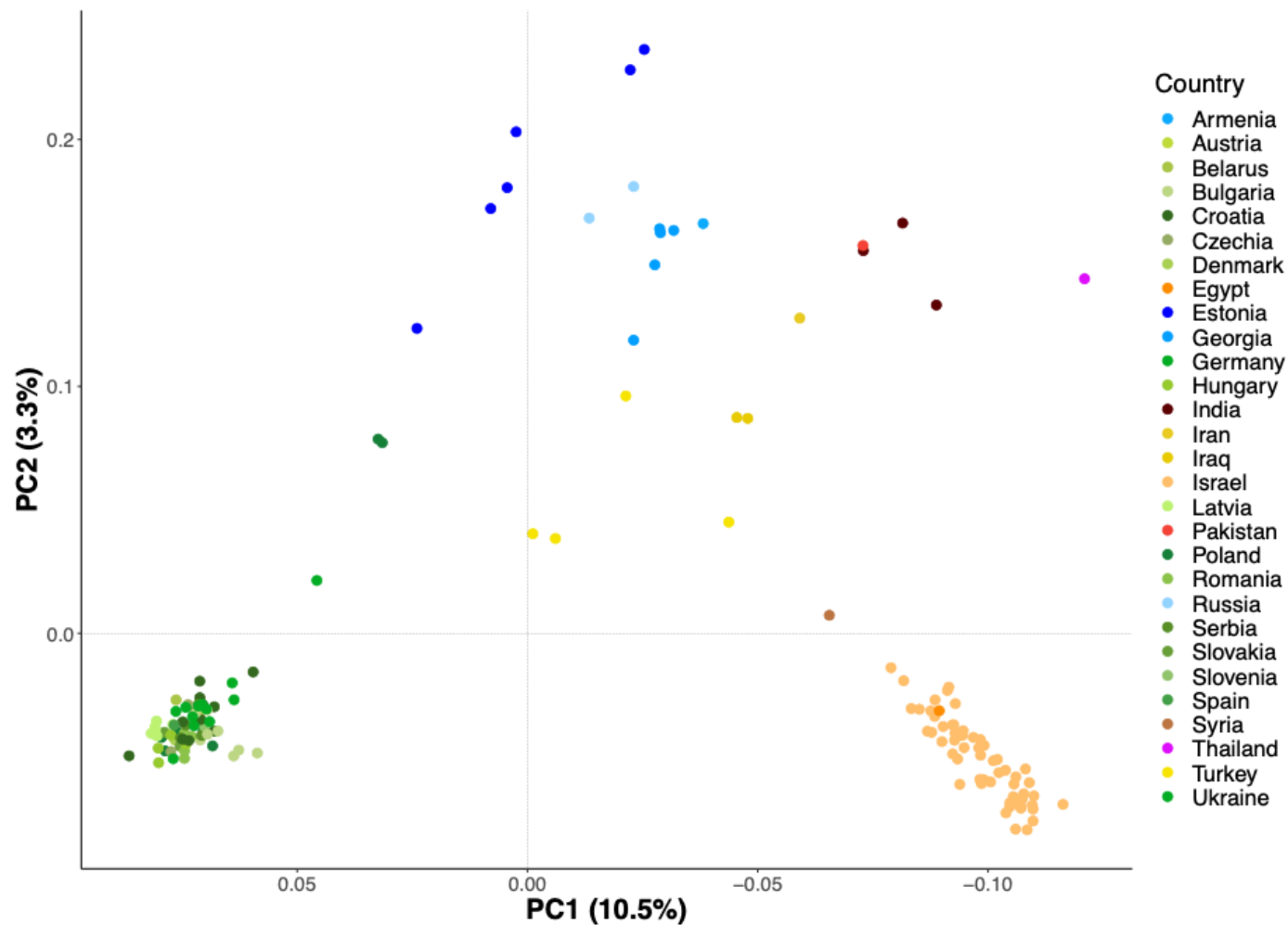
Kinship 0.48

PCA results – population structure

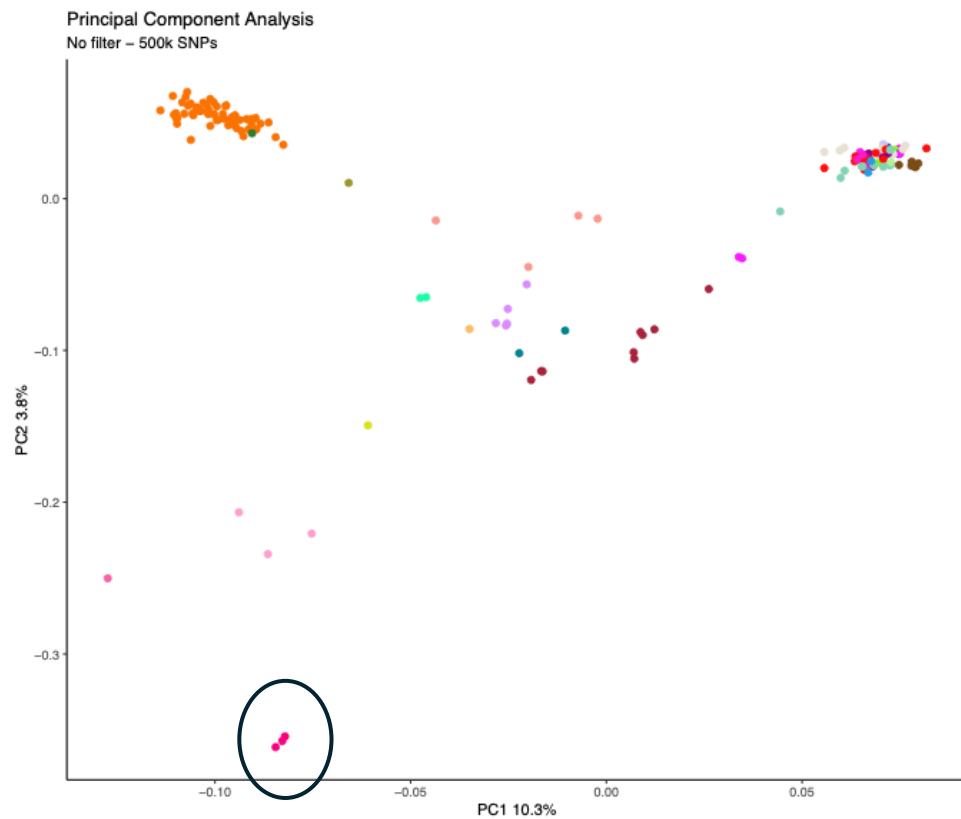


Pakistan

Kinship 0.48

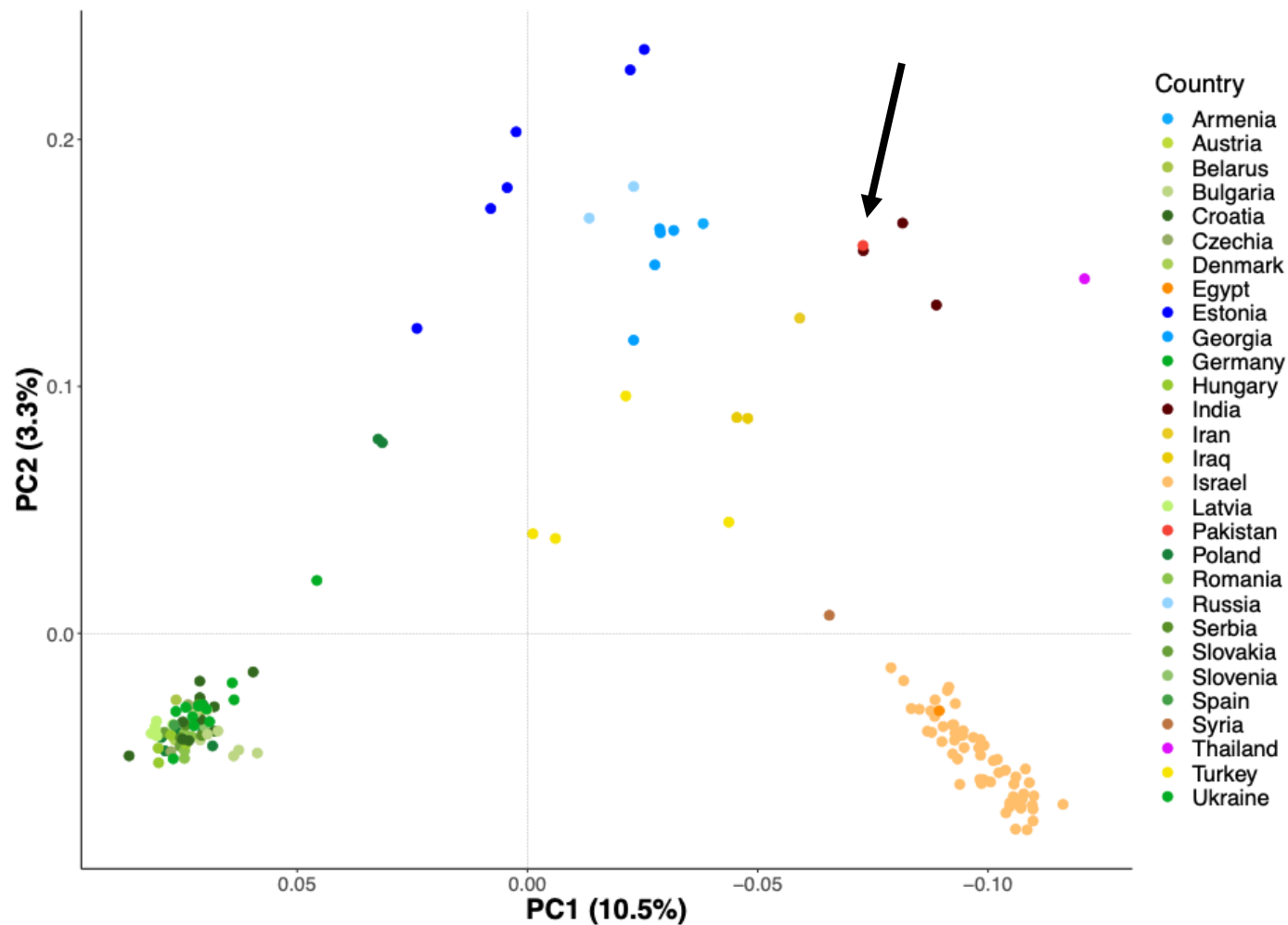


PCA results – population structure



Pakistan

Kinship 0.48



Relatedness is important

1

PCA results - population structure

Many analysis assume that all individuals analyzed are unrelated and if related individuals are not removed or controlled for they can lead to highly inflated false-positive rates.

2

Diseases

If affected relatives share a chromosomal segment **IBD** more often than expected by chance, that segment may contain a disease-causing variant.

3

Inbreeding

4

Conservation

Relatedness is important

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PCA results - population structure

Many analysis assume that all individuals analyzed are unrelated and if related individuals are not removed or controlled for they can lead to highly inflated false-positive rates.

2

Diseases

If affected relatives share a chromosomal segment **IBD** more often than expected by chance, that segment may contain a disease-causing variant.

3

Inbreeding

IBD is the mechanism behind inbreeding: an individual is inbred at a locus when its two homologous alleles are **identical by descent** from the same ancestral allele.

High relatedness produces long runs of homozygosity (**ROH**)

4

Conservation

Relatedness is important

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PCA results - population structure

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High relatedness produces long runs of homozygosity (**ROH**)

4

Conservation

A pedigree-based management strategy that minimizes mean kinship is effective at **mitigating drift** and **inbreeding**, while preserving genetic diversity and evolutionary potential to curtail extinction risk

How to calculate it? Relatedness

King

relateAdmix

Li & Ro

KING – Kinship-based inference

KING-robust kinship estimator (Manichaikul et al. 2010)

$$\hat{\phi}_{ij} = \frac{[IBS2^* - 2 \cdot IBS0]}{[4 \cdot \min(HET_i, HET_j)]}$$

IBS0 = SNPs where both are homozygous but for different alleles

IBS2* = SNPs where both are heterozygous

HET = proportion of heterozygous sites per individual

Expected kinship thresholds

$$\phi > 0.354$$

Duplicate / MZ twin

$$0.177 - 0.354$$

1st degree (parent/offspring, full sib)

$$0.088 - 0.177$$

2nd degree (half-sib, grandparent, uncle)

$$0.044 - 0.088$$

3rd degree (1st cousins)

$$\phi < 0.044$$

Unrelated

KING-robust handles global structure but it will be biased in admixed samples.

Most earlier estimators (e.g., PLINK --genome) assume random mating and break down across diverged populations. KING-robust is approximately unbiased even for between-population pairs, because IBS0 acts as a correction for allele frequency divergence.

Limitation: biased with strong population structure. For structured datasets → use relateAdmix or PC-Relate.

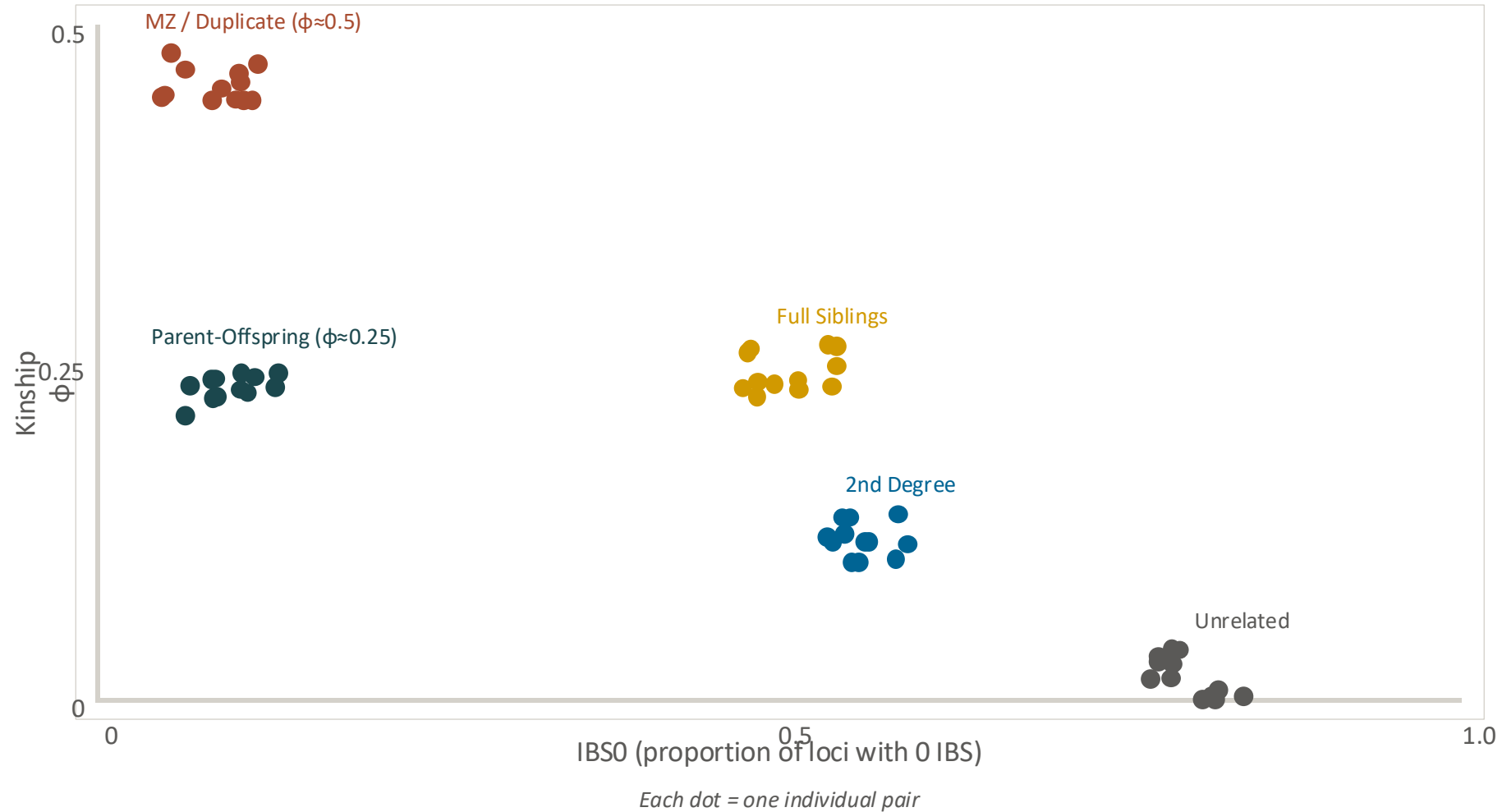
KING in PLINK 2

```
# PLINK 2 KING-robust
plink2 --bfile dataset \
  --make-king-table \
  --king-table-filter 0.044 \
  --out kinship_results

# Columns: ID1 ID2 NSNP HETHET IBS0 KINSHIP
```

Relationship Categories: Kinship vs IBS0

KING-robust classifies pairs using kinship coefficient and IBS0 proportion



relateAdmix: Why ancestry matters

The Problem

Standard estimators, like PLINK, use average population allele frequencies as background.

Admixed individuals draw alleles from MULTIPLE ancestral populations—so the "background frequency" varies along the genome.

Using a single-population baseline inflates false kinship between admixed individuals from the same source populations.

KING has the advantage that it does not need more than two samples (does not use allele frequencies)

KING tries to handle global structure but is biased in admixed samples.

relateAdmix Solution

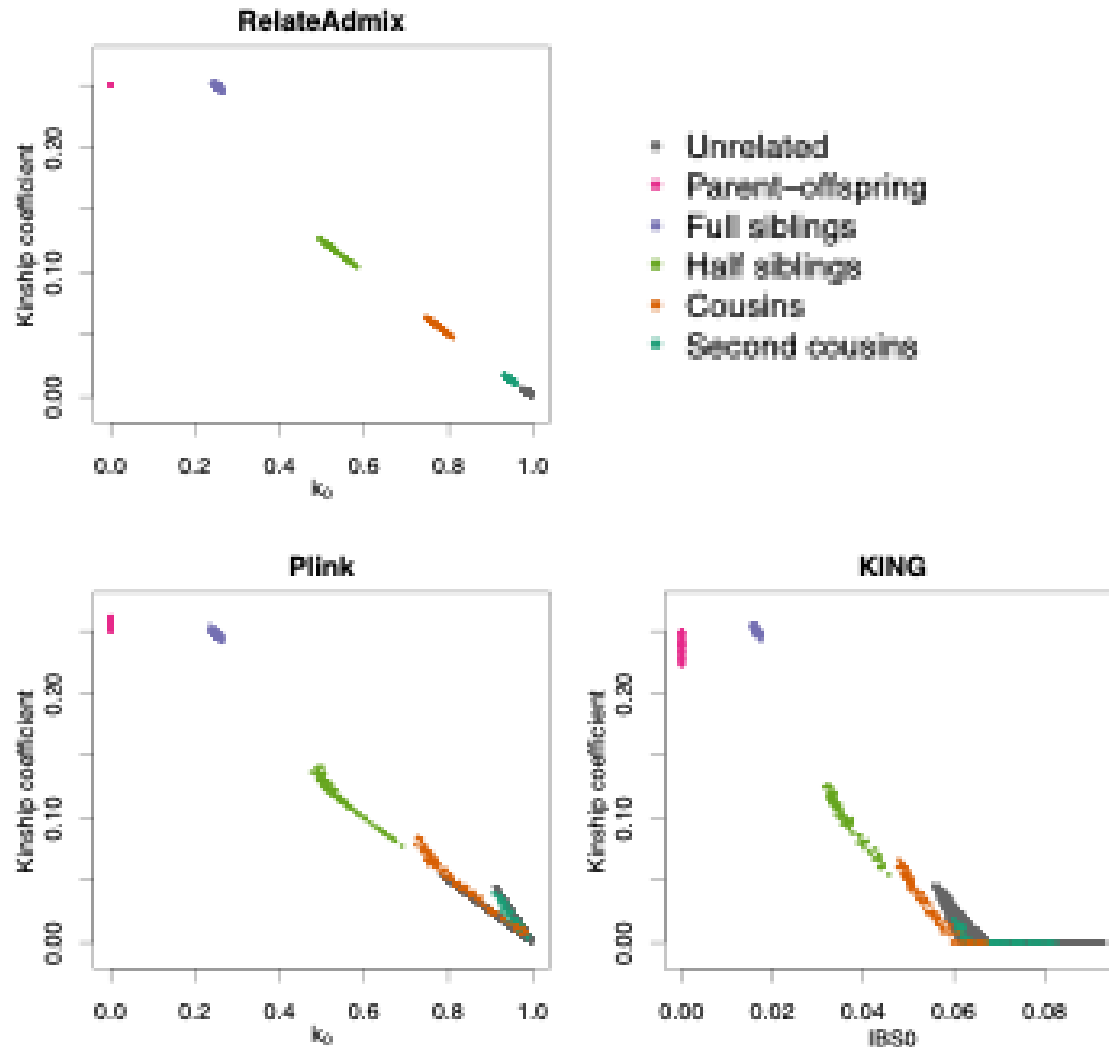
Decomposes allele sharing into:

1. Sharing due to IBD (recent common ancestry)
2. Sharing due to population ancestry

Uses **local ancestry** inference to estimate how much sharing is expected from ancestry alone, then subtracts this to get the IBD component.

Key advantage: Works in admixed populations where KING is biased.

Relatedness and structure



King does **not handles structure** perfectly.

We still use it, but we take each population separately

When we have **strong structure**, we use **RelateAdmix**

Key takeaways

1

Relatedness

IBD is the biological foundation. All relatedness estimators approximate the kinship coefficient ϕ —the **probability that two alleles are inherited from a common ancestor**.

Related individuals distort all downstream statistics leading to wrong conclusions

KING-robust is the recommended default pairwise estimator. Use relateAdmix for admixed populations, ngsRelate for low-coverage data.

2

Fst

Fst quantifies **population differentiation**. It can be defined as the ratio of the variance in allele frequencies among populations to the total genetic variance.

Low-diversity populations can show inflated FST even with minor allele frequency differences.

Population structure can inflate the Fst estimation

- Manichaikul et al. 2010 (KING paper)
- Moltke & Albrechtsen 2014 (relateAdmix paper)
- Korneliussen et al. 2014 (ANGSD paper)
- Weir & Cockerham 1984 (FST estimator—the canonical reference)

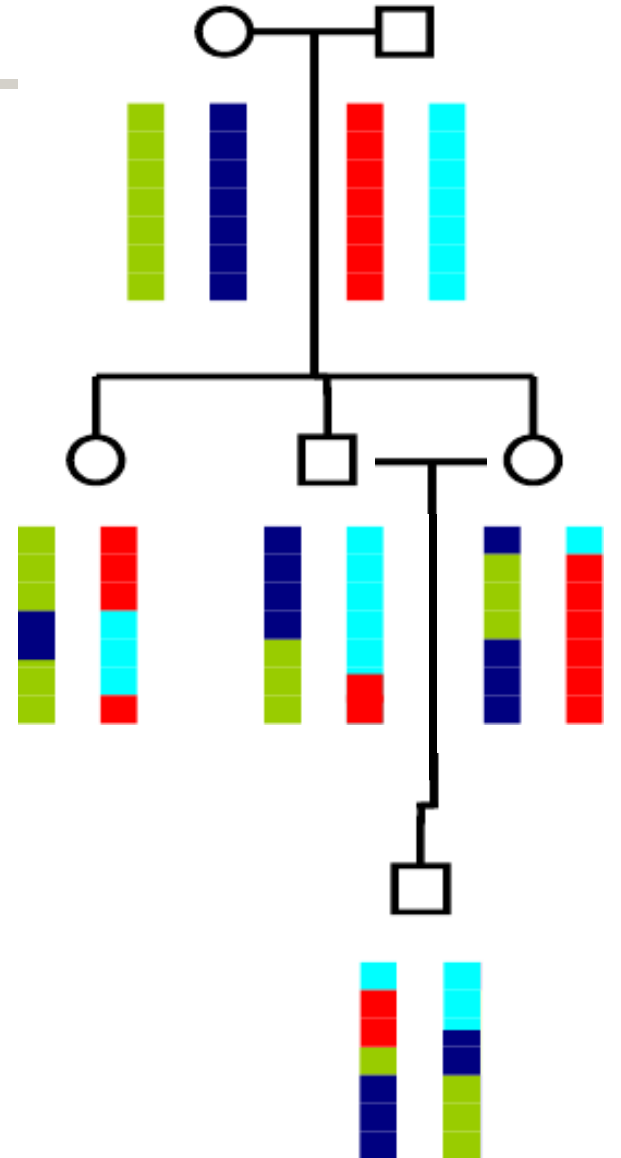
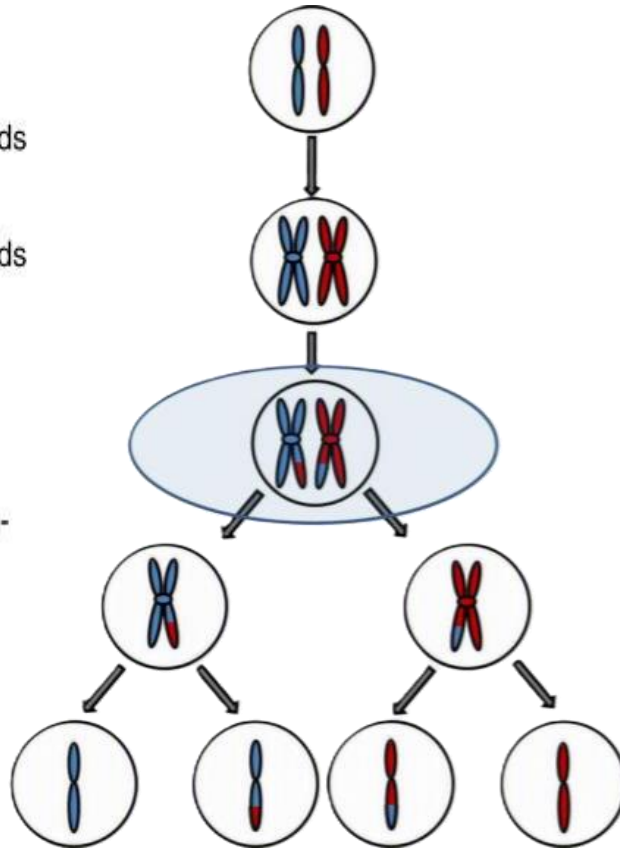
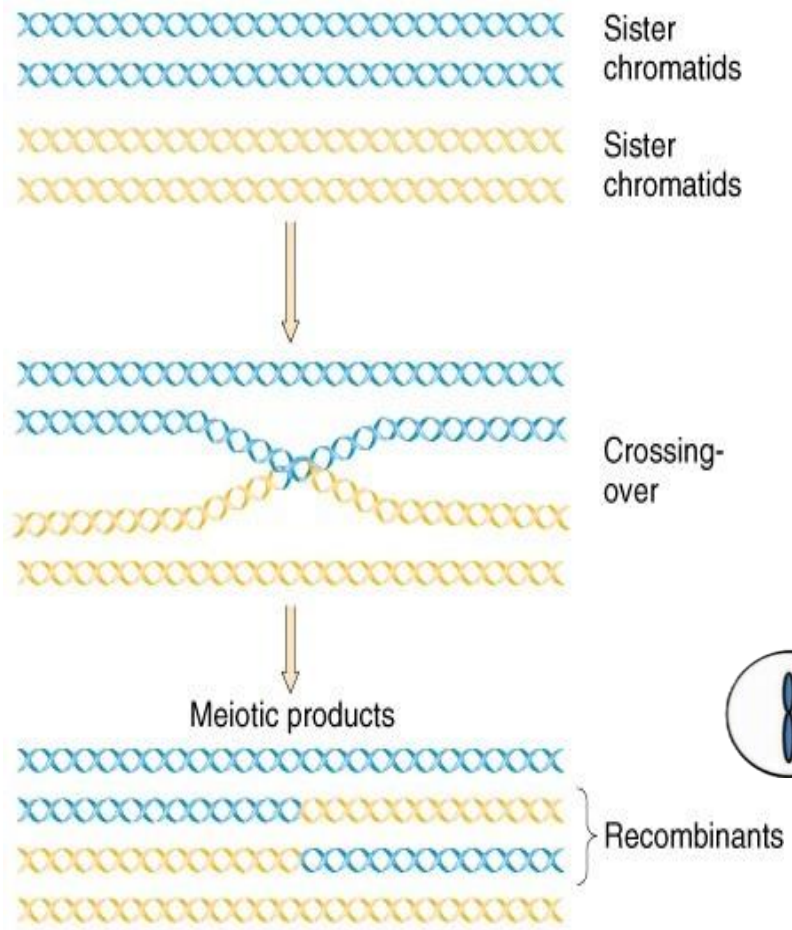
Time for exercises

Please do the Relatedness and stop before the
Fst in Wildebeest



Not so simple

Meiosis and recombination



relateAdmix – Relatedness in admixed populations

The problem with admixed samples

Standard estimators (PLINK, even KING) use average population allele frequencies as background. **Admixed individuals draw alleles from MULTIPLE ancestral populations**—so the "background frequency" varies along the genome.

Using a single-population baseline inflates false kinship between admixed individuals from the same source populations.

relateAdmix solution (Moltke & Albrechtsen 2014)

Uses an ADMIXTURE Q-matrix (individual ancestry proportions) + P-matrix (population allele frequencies) to compute individual-specific expected allele frequencies at each locus. Kinship is then estimated relative to this individual-tailored background.

Output: pairwise relatedness coefficients (ϕ) for all sample pairs

Workflow diagram

ADMIXTURE / NGSadmix
Q + P matrices



VCF / BEAGLE GL file
Input genotypes



relateAdmix
Admixture-corrected kinship



Pairwise ϕ matrix
Filter relatives

Pipeline

```
# 1. Run ADMIXTURE (K=3 example)
admixture --cv dataset.bed 3
# → dataset.3.Q dataset.3.P

# 2. Run relateAdmix
relateAdmix -plink dataset \
  -qfile dataset.3.Q \
  -pfile dataset.3.P \
  -out relatedness.txt

# Output: IND1 IND2 PHI
```